

A finite family with transcendental uniform Turán density

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Abstract

Uniform Turán density measures how dense a forbidden-hypergraph-free hypergraph can remain when density is tested uniformly over all lower-order structures. We construct a finite forbidden family whose top nontrivial uniform Turán density is transcendental. More precisely, for the explicit integer $K = 14,484,112$, we construct a finite family $\mathcal{F}$ of K -graphs and rational numbers α_0 and α_1 , with $\alpha_1 > 0$, such that

$$\pi_{K-2}^{(K)}(\mathcal{F}) = (\alpha_0 + \alpha_1 G(1/100))^2, \quad G(q) := \frac{1}{(1-q)(1-q^2)} \prod_{n \geq 1} (1-q^n)^{-1}.$$

Nesterenko's theorem implies that $G(1/100)$ is transcendental. The proof combines a cancellative ambient monoid and its 0-cancellative partition-growth Rees quotient, an exact quadratic realization of finite truncations, a stopped-tree estimate, and an effective conversion from typed positive constraints to untyped, unordered forbidden hypergraphs.

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1 Introduction

For an integer $K \geq 3$, a K -graph is a K -uniform hypergraph. Uniform Turán theory asks how dense a large K -graph may be when a fixed family of configurations is forbidden and the density condition is imposed on every test object of a prescribed lower uniformity. At the top nontrivial level, these test objects are $(K-2)$ -graphs, and we denote the corresponding extremal parameter by $\pi_{K-2}^{(K)}(\mathcal{F})$. The precise definition appears in [Section 2](#). This line of research originates in the uniformly dense problems of Erdős and Sós [\[5\]](#). Random color-pattern constructions underlying the modern palette formalism already appear in Rödl's work [\[29\]](#), while the top-level density belongs to the hierarchy introduced by Reiher, Rödl, and Schacht [\[27\]](#). Reiher [\[25\]](#) gives a broader overview.

Several exact results for a single forbidden hypergraph provide useful context. Glebov, Král', and Volec [\[9\]](#) and, independently, Reiher, Rödl, and Schacht [\[28\]](#) proved that the broken tetrahedron has uniform Turán density $1/4$. Reiher, Rödl, and Schacht [\[26\]](#) subsequently characterized the single K -graphs of zero top-level density and showed that every positive value is at least K^{-K} . This lower endpoint was attained for $K = 3$ by Garbe, Král', and Lamaison [\[8\]](#), and for every $K \geq 3$ by Lin, Wang, and Zhou [\[19\]](#). Further exact values are known for tight cycles [\[3\]](#) and for other small 3-graphs with densities $1/4$ and $8/27$ [\[4, 7\]](#).

A finite *palette* specifies a color set and a collection of admissible ordered K -tuples of colors. Its Lagrangian is the largest total weight of the admissible tuples over all probability weights on the colors. Lamaison [\[16\]](#) proved the palette formula for individual 3-graphs, Král', Kučerák, Lamaison, and Tardos [\[15\]](#) developed a palette classification theorem, and King, Piga, Sales, and Schülke [\[11\]](#) realized every 3-palette Lagrangian as the uniform Turán density of a finite forbidden family. Lin, Sun, Wang, and Zhou [\[18, Theorem 1.4\]](#) proved that, for every $K \geq 3$ and every forbidden family $\mathcal{F}$, the density $\pi_{K-2}^{(K)}(\mathcal{F})$ is the supremum of the densities of all finite palettes under which no member of $\mathcal{F}$ is colorable.

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Arithmetic questions have a longer history for ordinary hypergraph Turán density. Baber and Talbot [1, Theorem 22] found three forbidden 3-graphs whose joint Turán density is irrational. Pikhurko [24, Theorems 2 and 5] proved that finite forbidden families have irrational densities in every uniformity at least 3 and that the spectrum for arbitrary forbidden families has the cardinality of the continuum. Grosu [10] developed further algebraic and topological structure and obtained explicit irrational values, while Liu and Pikhurko [20, Theorem 1.2] constructed finite families whose densities have arbitrarily large algebraic degree.

The palette formula raises the analogous arithmetic question for uniform Turán density. The Lagrangian of each fixed finite palette is algebraic: quantifier elimination over real closed fields, equivalently the Tarski–Seidenberg theorem [2, Theorem 2.2.1], applies to its rational semialgebraic optimization problem. A finite forbidden family, however, controls infinitely many deficient palettes, and a supremum of algebraic Lagrangians need not be algebraic.

At the level of spectra, King, Sales, and Schülke [12, Theorem 1.1] proved that the uniform Turán densities of arbitrary families of 3-graphs are not well ordered. Li, Liu, and Pikhurko [17] proved that this spectrum contains a terminal interval, which in particular yields transcendental values. Liu and Pikhurko [21] proved that the uniform Turán densities of finite families of 3-graphs are dense in a nondegenerate interval. Density alone does not determine the arithmetic nature of any one value, and none of these results supplies a finite forbidden family whose density is proved transcendental. We construct such a family, with explicit uniformity.

Theorem 1.1. *Let*

$$\gamma := G(1/100), \quad G(q) := \frac{1}{(1-q)(1-q^2)} \prod_{n \geq 1} (1-q^n)^{-1}.$$

For $K = 14,484,112$, there exist an effectively constructible finite family $\mathcal{F}$ of K -graphs and rational numbers α_0 and α_1 , with $\alpha_1 > 0$, such that

$$\pi_{K-2}^{(K)}(\mathcal{F}) = (\alpha_0 + \alpha_1 \gamma)^2. \quad (1.1)$$

The value of K is not intended to be close to optimal. It makes the divisibility, separation, and monotonicity conditions in the homogeneous construction simultaneous and explicit. Here *effectively constructible* has a literal meaning: a terminating deterministic algorithm outputs adjacency-list encodings of all members of $\mathcal{F}$. The ancillary files give a complete machine-readable expansion of the typed signature and query manifest, together with an independently checkable rational certificate for the homogeneous parameters. The signal words and the final forbidden hypergraphs are specified by terminating deterministic rules; they are not enumerated in the ancillary files.

The transcendence of γ follows from Nesterenko’s algebraic independence theorem. Thus the exact formula immediately has the following consequences.

Corollary 1.2. *There exist an integer $K \geq 4$ and a finite family $\mathcal{F}$ of K -graphs such that $\pi_{K-2}^{(K)}(\mathcal{F})$ is transcendental.*

Equivalently, the top nontrivial uniform Turán densities of finite forbidden families are not all algebraic numbers.

Corollary 1.3. *For some integer K and some finite family $\mathcal{F}$ of K -graphs, the supremum of $\Lambda(P)$ over all finite $\mathcal{F}$ -deficient palettes P is transcendental.*

We briefly describe the proof. Its arithmetic source is Kobayashi’s monoid with zero [13]; following Koçak [14], we present it as the Rees quotient of the cancellative monoid $N_0 = \text{Mon}\langle a, b, c \mid ba = ab, bc = ac \rangle$ by the ideal generated by acc . The quotient is 0-cancellative, its surviving normal forms are indexed by integer partitions, and its weighted growth series is $G(q)$. Nesterenko’s theorem then gives the transcendental limit $\gamma = G(1/100)$.

The second ingredient is an exact finite optimization. Every finite rooted truncation of the normal-form tree is represented by two nonnegative vector fields. A completed-square identity computes the optimum and forces the correct rational edge ratios. These ratios can be expressed by finitely many positive typed relations. The central estimate examines the first place at which a proposed record ceases to follow a legal normal-form path. Local failures are charged to bounded cells, while the first entry into the Rees

ideal is controlled by a canonical descent and one Schur-type column estimate. The resulting bound gives a finite weighted system whose optimum converges to γ and whose error is uniformly controlled.

The final step removes the auxiliary language. We assign every signal the same multiset of atomic types, homogenize all coefficients, and encode the positive queries by ordered hypergraph gadgets. Pinned lifts preserve the intended identifications, while an outer-ordering construction removes the chosen vertex order. Applying the palette-transfer theorem to the resulting untyped, unordered forbidden family gives the upper bound. The intended palettes give the matching lower bound, and the two coordinate halves produce the square in (1.1).

The paper is organized as follows. Section 2 fixes the uniform-density conventions and proves the weighted palette transfer. Section 3 constructs the partition-growth monoid, proves the transcendence input, and establishes the completed-square identity. Section 4 builds the finite positive system and proves the stopped-tree estimate. Section 5 transfers this system to untyped, unordered K -graphs and proves the exact density formula. The appendices, Sections A and B, give the exact typed query manifest, the rational parameter certificate, and the deterministic construction algorithm.

We write $\mathbb{N} := \{0, 1, 2, \dots\}$ and use standard asymptotic notation. Whenever the construction requires a finite choice, order, or search, we use the fixed deterministic convention specified in the relevant section.

2 Uniform density and palette transfer

We first fix the top-level uniform-density conventions and recall the palette transfer that converts the extremal problem into a weighted palette optimization.

A K -graph is a K -uniform hypergraph. For a $(K-2)$ -graph J , let $\mathcal{K}_K(J)$ be the family of K -subsets of $V(J)$ that span complete $(K-2)$ -graphs. A K -graph H on n vertices is $(d, \mu, K-2)$ -dense if

$$|E(H) \cap \mathcal{K}_K(J)| \geq d|\mathcal{K}_K(J)| - \mu n^K$$

for every $(K-2)$ -graph J on $V(H)$. A K -graph is $\mathcal{F}$ -free if it contains no copy of any member of $\mathcal{F}$. For a family $\mathcal{F}$ of K -graphs, set

$$\pi_{K-2}^{(K)}(\mathcal{F}) := \sup\{d \in [0, 1] : \text{for every } \mu > 0 \text{ and } n_0 \in \mathbb{N}, \text{ there is an } \mathcal{F}\text{-free } (d, \mu, K-2)\text{-dense } K\text{-graph on at least } n_0 \text{ vertices}\}.$$

Here and below the supremum of an empty subset of $[0, 1]$ is understood to be 0.

A K -palette P consists of a finite nonempty color set $C(P)$ and a set $A(P) \subseteq C(P)^K$ of admissible ordered tuples. Its unweighted density is $d(P) := |A(P)|/|C(P)|^K$. For a probability vector $w = (w_c)_{c \in C(P)}$, define

$$d_w(P) := \sum_{(c_1, \dots, c_K) \in A(P)} w_{c_1} \cdots w_{c_K}, \quad \Lambda(P) := \max_w d_w(P).$$

A K -graph F is P -colorable if one can choose a linear order on $V(F)$ and a coloring ϕ of the covered $(K-1)$ -shadow of F by $C(P)$ so that, for every edge $e = \{v_1 < \dots < v_K\}$, the ordered tuple

$$(\phi(e \setminus \{v_1\}), \dots, \phi(e \setminus \{v_K\}))$$

belongs to $A(P)$. Because $C(P)$ is nonempty, this is equivalent to coloring the full $(K-1)$ -shadow. Colors on uncovered faces are irrelevant and may be chosen arbitrarily. A palette is $\mathcal{F}$ -deficient if no member of $\mathcal{F}$ is P -colorable. The *reverse palette* is

$$\text{rev}(P) := (C(P), \{(c_K, \dots, c_1) : (c_1, \dots, c_K) \in A(P)\}).$$

A *palette homomorphism* from P to Q is a map from $C(P)$ to $C(Q)$ that preserves every admissible tuple. A palette is *nondegenerate* if every admissible tuple has pairwise distinct colors.

Lemma 2.1. *The definitions above agree with the conventions of Lin, Sun, Wang, and Zhou [18, Definitions 1.1–1.3]. In particular, their palette Turán density of a family $\mathcal{F}$ is*

$$\sup\{d(P) : P \text{ is a finite } \mathcal{F}\text{-deficient palette}\}.$$

Proof. The two notions of $(d, \mu, K-2)$ -density are identical. The cited paper colors the full $(K-1)$ -shadow, whereas we color only the covered shadow. Restricting a full-shadow coloring gives one implication. Conversely, because the color set is nonempty, every coloring of the covered shadow extends to the uncovered faces. Finally, a family is palette-colorable in the cited convention precisely when at least one of its members is palette-colorable. Thus failure of family colorability is exactly the notion of deficiency used here. $\square$

Theorem 2.2 (Lin–Sun–Wang–Zhou [18, Theorem 1.4]). *For every $K \geq 3$ and every family $\mathcal{F}$ of K -graphs,*

$$\pi_{K-2}^{(K)}(\mathcal{F}) = \sup\{d(P) : P \text{ is a finite } \mathcal{F}\text{-deficient palette}\}.$$

This theorem is the only external uniform-density transfer used below; the classification theorem of Král' et al. [15] and the realization theorem of King et al. [11] are used only for context. The interface in Lemma 2.1 isolates all convention-dependent points.

We use the following weighted form.

Lemma 2.3. *Let P be a finite palette equipped with a probability vector w . There exist finite palettes P_n and homomorphisms $P_n \rightarrow P$ such that every tuple of P_n has pairwise distinct colors and $d(P_n) \rightarrow d_w(P)$. If P is $\mathcal{F}$ -deficient, then every P_n is $\mathcal{F}$ -deficient.*

Proof. For each n , choose integers $N_c(n) \geq 0$ such that $\sum_c N_c(n) = n$ and $N_c(n)/n \rightarrow w_c$. Replace c by $N_c(n)$ clones, pull back every admissible tuple, and discard clone tuples with a repeated clone. For an original tuple, let q_c be the multiplicity of the color c . Its contribution to the normalized density is $n^{-K} \prod_{c \in C(P)} (N_c(n))_{q_c}$, where $(m)_q := m(m-1) \cdots (m-q+1)$ and $(m)_0 := 1$. Since $n^{-q_c} (N_c(n))_{q_c} \rightarrow w_c^{q_c}$ for every c , summing over the finitely many original tuples gives $d(P_n) \rightarrow d_w(P)$. Projection onto the original colors is a palette homomorphism. By construction every admissible tuple of P_n has pairwise distinct colors, so P_n is nondegenerate. Composing a P_n -coloring with this projection proves that deficiency is preserved. $\square$

Corollary 2.4. *For every family $\mathcal{F}$ of K -graphs,*

$$\pi_{K-2}^{(K)}(\mathcal{F}) = \sup\{\Lambda(P) : P \text{ is a finite } \mathcal{F}\text{-deficient palette}\}.$$

Proof. The right side is at least the unweighted palette supremum. Conversely, apply Lemma 2.3 to an arbitrary weighting of a deficient palette and then use Theorem 2.2. $\square$

3 The partition-growth source

This section constructs the arithmetic source of the proof. We first obtain partition growth from a cancellative monoid and its Rees quotient, then apply Nesterenko's theorem to the limiting series, and finally realize each finite normal-form tree by an exact quadratic maximization.

3.1 A cancellative source with partition growth

The transcendental number in Theorem 1.1 enters through a weighted growth series. The Rees quotient used by Kobayashi [13] has presentation $\text{Mon}\langle a, b, c \mid ba = ab, bc = aca, acc = 0 \rangle$; see also Koçak [14, Section 6]. We first isolate the cancellative precursor obtained by omitting the zero relation. Its two relations move every letter b to the right and turn the surviving blocks into a nondecreasing sequence, which is precisely where integer partitions enter the construction.

Let $N_0 := \text{Mon}\langle a, b, c \mid ba = ab, bc = aca \rangle$, and give the generators the positive grading $\deg(a) := 1$, $\deg(c) := 1$, and $\deg(b) := 2$. Both defining relations are homogeneous of degree three. Orient them as

$$ba \longrightarrow ab, \quad bc \longrightarrow aca. \tag{3.1}$$

Lemma 3.1. *The rewriting system (3.1) is terminating and confluent. Every element of N_0 has a unique normal form $w(a, c)b^k$, where $w \in \{a, c\}^*$ and $k \geq 0$.*

Proof. For a word w , let $b(w) := |\{i: w_i = b\}|$ and let $\iota(w) := |\{(i, j): i < j, w_i = b, w_j = a\}|$. Order words lexicographically by $(b(w), \iota(w))$. The rule $bc \rightarrow aca$ strictly decreases $b(w)$, while $ba \rightarrow ab$ preserves $b(w)$ and strictly decreases $\iota(w)$. Thus no infinite reduction sequence exists. The left sides ba and bc have no overlap, so the system is locally confluent. Newman's lemma [23] now implies confluence. A reduced word has no a or c to the right of a b , and hence has the displayed form. Its uniqueness follows from confluence. $\square$

Lemma 3.2. *The monoid N_0 is left and right cancellative.*

Proof. Write an element in normal form as wb^k , with $w \in \{a, c\}^*$. Right multiplication by b gives wb^{k+1} . Right multiplication by a gives wab^k , because $b^ka = ab^k$. Both maps are injective. Moreover, $b^kc = a^kca^k$ by induction from $bc = aca$, and hence $(wb^k)c = wa^kca^k$. The rightmost c , followed by the terminal run a^k , recovers k , and deleting the preceding a^k recovers w . Hence right multiplication by c is injective.

Left multiplication by a and by c is immediate from the normal form. For left multiplication by b , either $w = a^r$, in which case $b(a^rb^k) = a^rb^{k+1}$, or $w = a^rcu$, where this is the first occurrence of c , in which case

$$b(a^rcu)b^k = a^{r+1}caub^k. \quad (3.2)$$

The two cases are distinguished by the presence of a c . In the second case, the number of letters a before the first c recovers $r + 1$, and hence r . Deleting the first a after this c then recovers u . Thus left multiplication by every generator is injective. Since multiplication by an arbitrary element is a composition of multiplications by generators, N_0 is left and right cancellative. $\square$

Let $\mathfrak{J} := N_0(acc)N_0$. For a word $w \in \{a, c\}^*$, write

$$w = a^{m_1}ca^{m_2}c \cdots a^{m_r}ca^\ell. \quad (3.3)$$

Here $r, \ell, m_1, \dots, m_r \in \mathbb{N}$; when $r = 0$, the displayed product is empty and $w = a^\ell$.

Proposition 3.3. *A normal form wb^k , with w written as in (3.3), lies outside $\mathfrak{J}$ if and only if $0 \leq m_1 \leq m_2 \leq \cdots \leq m_r$. Consequently the Rees quotient $N_0/\mathfrak{J}$ is 0-cancellative.*

Proof. Let $\mathcal{D} := \{wb^k: (m_1, \dots, m_r) \text{ has a descent}\}$. We first prove $\mathcal{D} \subseteq \mathfrak{J}$. If $m_i > m_{i+1}$, then put $d := m_i - m_{i+1} \geq 1$. Using the defining relations in both directions,

$$\begin{aligned} a^{m_i}ca^{m_{i+1}}c &= a^d(a^{m_{i+1}}ca^{m_{i+1}})c \\ &= a^db^{m_{i+1}}cc = b^{m_{i+1}}a^{d-1}(acc). \end{aligned}$$

Thus the corresponding element belongs to the two-sided ideal generated by acc .

Conversely, $\mathcal{D}$ is a two-sided ideal. Right multiplication by b leaves the sequence (m_i) unchanged, right multiplication by a changes only the terminal exponent ℓ , and right multiplication by c appends $\ell + k$ to the sequence, so an existing descent remains. Left multiplication by a increases m_1 ; left multiplication by c prepends zero. Finally, if a c occurs, (3.2) increases the first two block exponents by one, or increases the first block exponent and the tail when there is only one c ; every existing descent remains. Hence $\mathcal{D}$ is a two-sided ideal. Since $acc \in \mathcal{D}$, we have $\mathfrak{J} \subseteq \mathcal{D}$. The two inclusions prove the characterization.

Let $\bar{x}, \bar{y}$ be nonzero quotient elements. If $\bar{x}\bar{s} = \bar{y}\bar{s} \neq 0$ for a generator s , then the products xs and ys in N_0 lie outside $\mathfrak{J}$ and are equal. Right cancellation in N_0 , supplied by Lemma 3.2, gives $x = y$, hence $\bar{x} = \bar{y}$. The same argument using left cancellation proves left 0-cancellation. To cancel an arbitrary nonzero quotient element, choose a representing word $z = s_1 \cdots s_m$. If $\bar{x}\bar{z} = \bar{y}\bar{z} \neq 0$, then every intermediate product $\bar{x}\bar{s}_1 \cdots \bar{s}_j$ and $\bar{y}\bar{s}_1 \cdots \bar{s}_j$ is nonzero. Indeed, once an intermediate product entered the Rees ideal, ideal absorption would make the final product zero. One may therefore cancel $s_m, s_{m-1}, \dots, s_1$ successively. The left case is identical. $\square$

Proposition 3.4. *With $\mathcal{P}(q) := \prod_{n \geq 1} (1 - q^n)^{-1}$, the exact growth series, as an identity in $\mathbb{Q}[[q]]$, is*

$$G(q) = \sum_{x \in N_0 \setminus \mathfrak{J}} q^{\deg(x)} = \frac{\mathcal{P}(q)}{(1-q)(1-q^2)}. \quad (3.4)$$

Both sides converge analytically when $|q| < 1$.

Proof. The degree of a block $a^m c$ is $m + 1$. By [Proposition 3.3](#), the sequence of positive integers $m_1 + 1 \leq \dots \leq m_r + 1$ is an integer partition, whose generating function is $\mathcal{P}(q)$. The empty block sequence $r = 0$ corresponds to the empty partition and supplies the constant term 1. The terminal factors a^ℓ and b^k contribute $(1 - q)^{-1}$ and $(1 - q^2)^{-1}$, respectively. The uniqueness in [Lemma 3.1](#) makes the product exact. $\square$

3.2 Transcendence of the growth value

We next show that the growth value at $q = 1/100$ is transcendental. The only number-theoretic input is Nesterenko's algebraic-independence theorem for the Eisenstein series. We use the following standard normalizations.

For $j \geq 0$, let $\sigma_j(n) := \sum_{d|n} d^j$. For τ in the upper half-plane and $q := e^{2\pi i \tau}$, define the normalized Eisenstein series

$$E_2(q) := 1 - 24 \sum_{n \geq 1} \sigma_1(n) q^n, \quad E_4(q) := 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n, \quad E_6(q) := 1 - 504 \sum_{n \geq 1} \sigma_5(n) q^n,$$

and the modular discriminant $\Delta(q) := q \prod_{n \geq 1} (1 - q^n)^{24}$.

Theorem 3.5 (Nesterenko [\[22, Theorem 1\]](#)). *Let $q := e^{2\pi i \tau}$, where τ lies in the upper half-plane. Then*

$$\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(q, E_2(q), E_4(q), E_6(q)) \geq 3.$$

Corollary 3.6. *If q is nonzero algebraic and $|q| < 1$, then $\mathcal{P}(q)$ and $G(q)$ are transcendental. In particular, $G(1/100)$ is transcendental.*

Proof. Since q is algebraic, [Theorem 3.5](#) implies that $E_2(q), E_4(q), E_6(q)$ are algebraically independent over $\mathbb{Q}$. Since $\overline{\mathbb{Q}}/\mathbb{Q}$ is algebraic, these three values remain algebraically independent over $\overline{\mathbb{Q}}$. The identity [\[22, p. 1320\]](#)

$$\Delta(q) = \frac{E_4(q)^3 - E_6(q)^2}{1728} = q \mathcal{P}(q)^{-24}$$

shows that, if $\mathcal{P}(q)$ were algebraic, then $\Delta(q) = q \mathcal{P}(q)^{-24}$ would lie in $\overline{\mathbb{Q}}$ and $E_4(q)^3 - E_6(q)^2 - 1728 \Delta(q) = 0$ would be a nonzero polynomial relation over $\overline{\mathbb{Q}}$, a contradiction. The Euler product defining $\mathcal{P}(q)$ converges to a nonzero value for $|q| < 1$. The two remaining factors in [\(3.4\)](#) are nonzero algebraic numbers, so multiplication by them does not change transcendence. $\square$

3.3 The finite two-field source

The growth series is an infinite sum, whereas the forbidden family must be finite. We therefore work first with a finite rooted truncation. The next identity realizes the squared path-weight sum of such a tree as the exact maximum of a quadratic functional in two nonnegative fields. The second field converts the desired linear edge ratios into positive local constraints of the correct scale; this feature is essential for the positive-query implementation used here.

Let T be a finite rooted tree with root o . Every edge has a label s from a finite alphabet Σ and a positive rational weight r_s . Define

$$B_T \delta_x := \sum_{x \xrightarrow{s} z} r_s \delta_z, \quad C_T := \text{Id} - B_T.$$

We regard vectors as columns. Thus the column of B_T indexed by x records the weighted children of x , while the row indexed by a nonroot vertex records its unique parent. For the unique path from o to x , let $r(x)$ be the product of its edge weights. Since B_T is nilpotent,

$$h_T := C_T^{-1} \delta_o = \sum_{x \in T} r(x) \delta_x, \quad g_T := \|h_T\|_2^2 = \sum_{x \in T} r(x)^2. \quad (3.5)$$

For $y \geq 0$ and nonnegative vectors u and v in $\mathbb{R}^{V(T)}$, put

$$\mathcal{J}_T(y, u, v) := 2yv_o - \|u\|_2^2 - \|u - C_T^\top v\|_2^2. \quad (3.6)$$

Theorem 3.7. *For every finite rooted weighted tree,*

$$\begin{aligned} \mathcal{J}_T(y, u, v) &= 2y^2 g_T - \frac{1}{2} \|C_T^\top v - 2yh_T\|_2^2 \\ &\quad - 2 \left\| u - \frac{1}{2} C_T^\top v \right\|_2^2. \end{aligned} \quad (3.7)$$

Consequently, $\max_{u, v \geq 0} \mathcal{J}_T(y, u, v) = 2y^2 g_T$. The unique optimizer is

$$u^* = yh_T, \quad v^* = 2yC_T^{-\top} h_T. \quad (3.8)$$

In particular, $u_{xs}^* = r_s u_x^*$ on every edge $x \xrightarrow{s} xs$.

Proof. Write $z := C_T^\top v$. Since $v_o = \langle C_T^{-1} \delta_o, z \rangle = \langle h_T, z \rangle$, expansion of the right side of (3.7) gives $2yv_o - 2\|u\|_2^2 + 2\langle u, z \rangle - \|z\|_2^2$, which is (3.6). The inverse matrices are finite sums of nonnegative matrices. Hence the vectors in (3.8) are feasible, and they make both squares vanish. The ratio assertion follows from (3.5). $\square$

We now specialize to $\Sigma = \{a, b, c\}$ by taking $t := 1/10$ and setting $r_a := t$, $r_c := t$, and $r_b := t^2$. For later use, put $r_{\max} := \max\{r_a, r_b, r_c\} = 1/10$. The two aggregate ratios are

$$r_1 := r_a + r_b + r_c = \frac{21}{100}, \quad r_2 := r_a^2 + r_b^2 + r_c^2 = \frac{201}{10000}. \quad (3.9)$$

For a word $q = s_1 \cdots s_m \in \{a, b, c\}^*$, write $r(q) := \prod_{j=1}^m r_{s_j}$, with $r(\emptyset) := 1$. The legal normal-form tree has one vertex for every normal word outside $\mathfrak{J}$. Before the first b , one may append a, c , or b ; after the first b , one may append only b . Retain an extension only when the resulting normal word lies outside $\mathfrak{J}$. Let T_D be the induced subtree of words of degree at most D .

Proposition 3.8. *Writing $g_D := g_{T_D}$, one has*

$$g_D \nearrow G(1/100) = \gamma. \quad (3.10)$$

Moreover, every finite rooted legal subtree satisfies

$$g_T \leq g_{\text{ub}} := \frac{1}{1 - r_2} = \frac{10000}{9799}. \quad (3.11)$$

Proof. A vertex represented by $x \in N_0 \setminus \mathfrak{J}$ has squared path weight $t^{2 \deg(x)}$. Thus (3.10) follows from Proposition 3.4 and monotone convergence. At distance j from the root, the total squared path weight is at most r_2^j . Summing over j gives (3.11). $\square$

The next finite models realize these trees exactly.

Definition 3.9. *A total-to-zero model consists of a finite set $X \cup \{0\}$, where $0 \notin X$, and maps R_a, R_b , and R_c from $X \cup \{0\}$ to itself such that $R_s(0) = 0$,*

$$R_a R_b = R_b R_a, \quad R_c R_b = R_a R_c R_a, \quad R_c R_c R_a \equiv 0,$$

and $R_s(x) = R_s(y) \neq 0$ implies $x = y$. Composition is read from right to left.

Proposition 3.10. *For every integer $D \geq 0$, there is a finite total-to-zero model in the sense of Definition 3.9 whose nonzero states are the elements of the Rees quotient $N_0/\mathfrak{J}$ of degree at most D . The scored tree formed by canonical normal-form transitions from the identity is T_D .*

Proof. Let X_D be the set of nonzero elements of $N_0/\mathfrak{J}$ of degree at most D . For $x \in X_D$ and $s \in \{a, b, c\}$, set

$$R_s(x) := \begin{cases} xs, & xs \neq 0 \text{ and } \deg(xs) \leq D, \\ 0, & \text{otherwise,} \end{cases} \quad R_s(0) := 0.$$

Consider one homogeneous defining relation $u = v$, namely $ba = ab$ or $bc = aca$. If $xu = xv$ is nonzero and has degree at most D , then no intermediate prefix on either side enters $\mathfrak{J}$, because $\mathfrak{J}$ is a right ideal, and no intermediate degree exceeds the common final degree. Both truncated paths therefore reach the

same nonzero state. If the common product is nonzero but has degree greater than D , homogeneity gives the same final degree on both sides, so both paths are truncated to zero. If the common product belongs to $\mathfrak{J}$, each path is eventually absorbed by the ideal and again ends at zero. Hence $R_a R_b = R_b R_a$ and $R_c R_b = R_a R_c R_a$. Right multiplication by acc enters $\mathfrak{J}$, so $R_c R_c R_a \equiv 0$. Finally, if $R_s(x) = R_s(y) \neq 0$, then $xs = ys$ outside $\mathfrak{J}$; 0-cancellation from [Proposition 3.3](#) gives $x = y$. The canonical normal-word transitions from the identity are precisely the edges of T_D . $\square$

It is worth recording why both fields are needed. If one tries to enforce the map directly in a single field, deleting an edge from a color of mass ξ to one of mass η is detected by a positive hereditary query only at scale $O(\xi\eta)$, whereas the completed-square deficit is of order ξ^2 . The semantic field u avoids this mismatch. At the optimizer its edge ratios are the fixed rational numbers r_a, r_b, r_c , and these ratios can be imposed by the finite positive packages constructed in the next section.

4 A finite positive system

The goal of this section is to construct a finite weighted relational system whose extremal value is governed by the finite normal-form trees in [Proposition 3.10](#), with their exact quadratic values supplied by [Theorem 3.7](#). The construction has three stages. We first encode the prescribed edge ratios by positive matching packages and a finite list of forbidden queries. We then prove a stopped-tree estimate. Every record that departs from a legal normal-form component pays a definite amount of the quadratic deficit. Finally, we choose the remaining positive coefficients so that the intended finite Rees models attain the matching value.

4.1 Typed records and positive queries

Definition 4.1. *Let $\mathcal{T}$ be a finite type set. A finite weighted typed record consists of pairwise disjoint finite color classes $(C_X)_{X \in \mathcal{T}}$, a nonnegative weight function on their union, and, for every directed relation symbol $R: X \rightarrow Y$, a relation $R \subseteq C_X \times C_Y$. A relation with $X = Y$ is required to be loopless. We write $a_X := \sum_{x \in C_X} w_x$ and $h(R) := \sum_{(x,y) \in R} w_x w_y$. A positive typed query is a finite conjunction of directed relation atoms, with a prescribed type for every variable. An occurrence is a type-preserving assignment of the variables to colors that satisfies all atoms and all written identifications. The assignment need not be injective, except where looplessness forces two images to be distinct. A record is query-free if none of the queries in the fixed finite list has an occurrence.*

The rest of the subsection builds the finite local language. We first express each prescribed ratio by a nonnegative matching defect. A balanced switch then converts all signed terms into positive relation densities. We finally list the finite queries that enforce functionality, cancellation, the defining relators, the Rees ideal, and the consistency between the two fields. The stopped-tree estimate is proved in [Section 4.2](#).

4.1.1 Ratio packages and the balanced switch

For a weighted finite set X , write $p_2(X) := \sum_{x \in X} w_x^2$ and $e_2(X) := \sum_{\{x,x'\} \in \binom{X}{2}} w_x w_{x'}$. Thus $p_2(X) + 2e_2(X) = a_X^2$, where $a_X := \sum_{x \in X} w_x$. Let X and Y be disjoint domain and codomain role copies. If $A \subseteq X \times Y$ is a partial matching, put $h(A) := \sum_{(x,y) \in A} w_x w_y$ and, for a fixed rational $r > 0$, define its *ratio defect* by

$$\Delta_r(A) := \frac{1}{2} (r^2 p_2(X) + p_2(Y)) - r h(A).$$

Lemma 4.2. *For a partial matching $A \subseteq X \times Y$,*

$$\begin{aligned} 2\Delta_r(A) = & \sum_{(x,y) \in A} (r w_x - w_y)^2 \\ & + r^2 \sum_{x \notin \text{dom } A} w_x^2 + \sum_{y \notin \text{ran } A} w_y^2. \end{aligned}$$

In particular, $\Delta_r(A) \geq 0$. Equality holds precisely when the restriction of A to the positive supports is a bijective matching covering both positive supports and $w_y = r w_x$ on every positive matched pair.

Proof. Expand the squares and use that a partial matching contains each endpoint at most once. $\square$

To make the same quantity visible to a positive palette score, attach to a binary relation $A \subseteq X \times Y$ directed loopless reserve relations $N_X \subseteq X^2$ and $N_Y \subseteq Y^2$. The finite manifest contains both orientations of every reserve choice. For this calculation, use the *normalized reserve density* $n(N) := \frac{1}{2} \sum_{(x,x') \in N} w_x w_{x'}$. When both orientations of every distinct pair are present, the normalized reserve density equals $e_2(X)$. Let $\text{Coll}_X(A)$ join two X -colors with a common A -neighbor, and define $\text{Coll}_Y(A)$ analogously. Collision queries require both orientations of every such pair to be absent from the corresponding reserve. Put

$$\widehat{\Delta}_r(A; N_X, N_Y) := \frac{1}{2}(r^2 a_X^2 + a_Y^2) - rh(A) - r^2 n(N_X) - n(N_Y),$$

Lemma 4.3. *If N_X and N_Y avoid the collision pairs of A , then $\widehat{\Delta}_r(A; N_X, N_Y) \geq 0$. After the support repair, when A is a partial matching and both reserves are complete on distinct positive pairs,*

$$\widehat{\Delta}_r(A; N_X, N_Y) = \Delta_r(A). \quad (4.1)$$

Proof. Define the unordered weighted collision densities by

$$c_X := \sum_{\{x,x'\} \in E(\text{Coll}_X(A))} w_x w_{x'}, \quad c_Y := \sum_{\{y,y'\} \in E(\text{Coll}_Y(A))} w_y w_{y'}.$$

We first prove

$$rh(A) \leq \frac{1}{2}(r^2 p_2(X) + p_2(Y)) + r^2 c_X + c_Y. \quad (4.2)$$

Write $\xi_x := r w_x$ for $x \in X$ and $\eta_y := w_y$ for $y \in Y$, and move all terms in (4.2) to the left. Suppose that x and x' form an edge of the X -collision graph. Vary ξ_x and $\xi_{x'}$ while keeping $\xi_x + \xi_{x'}$ fixed. The contribution $-\frac{1}{2}(\xi_x^2 + \xi_{x'}^2) - \xi_x \xi_{x'} = -\frac{1}{2}(\xi_x + \xi_{x'})^2$ is constant. Every other term is affine in the transfer parameter. Hence one endpoint of the transfer interval is no worse than the initial point. Moving to that endpoint removes one positive vertex from the collision edge. Repeating this operation leaves the positive X -support independent in its collision graph. Performing the same operation on Y does not change the X -support and leaves both positive supports independent. The restriction of A to these supports is then a partial matching. Summing $\xi_x \eta_y \leq \frac{1}{2}(\xi_x^2 + \eta_y^2)$ over its edges proves (4.2). Since each transfer did not decrease the left side minus the right side, the inequality also holds for the original weights.

The reserves avoid collision pairs, and therefore $n(N_X) \leq e_2(X) - c_X$ and $n(N_Y) \leq e_2(Y) - c_Y$. Combining these bounds with (4.2) and the identity $p_2(X) + 2e_2(X) = a_X^2$ proves nonnegativity. If A is a partial matching and the reserves are complete on distinct positive pairs, then $n(N_X) = e_2(X)$ and $n(N_Y) = e_2(Y)$. Substitution gives (4.1). $\square$

For each $s \in \{a, b, c\}$ introduce four auxiliary types O_s, D_s, Z_s, P_s and the following three relations.

- (i) a ratio matching $A_s: U \rightarrow O_s$ with ratio r_s ;
- (ii) an equality matching $K_s: U \rightarrow D_s$;
- (iii) an equality matching

$$T_s: O_s \dot{\cup} D_s \longrightarrow U \dot{\cup} Z_s \dot{\cup} P_s, \quad (4.3)$$

whose allowed type channels are $O_s \rightarrow U$, $O_s \rightarrow Z_s$, $D_s \rightarrow U$, and $D_s \rightarrow P_s$.

The aggregate type masses are constrained by $a_{O_s} = r_s a_U$, $a_{D_s} = a_U$, and $a_{Z_s} + a_{P_s} = r_s a_U$. Thus the two sides of (4.3) have equal total mass.

A semantic source $u \in U$ first follows A_s to a color $o \in O_s$. If $T_s(o) \in U$, then this target is the nonzero value $R_s(u)$. If $T_s(o) \in Z_s$, then the switch records the zero value. The D_s -copy associated with a target $z \in U$ is sent to z when $z \notin \text{ran } R_s$ and otherwise to its P_s -copy. Consequently a zero-defect switch represents a total map $R_s: U \cup \{0\} \rightarrow U \cup \{0\}$ with $R_s(0) = 0$.

The switch is used only after the phase and scored-role grading below. We therefore give the positive query manifest once, in its final graded form. The manifest enforces functionality, the allowed switch channels, the two monoid relations, the ideal relation, nonzero injectivity, source consistency, zero choices, and semantic-scored co-reference.

4.1.2 Graded scored roles and the positive ledger

We split all semantic types by the degree modulo three and all scored nonroot types by their incoming label. Thus every nonreserve relation joins two distinct atomic types, and the generic symbols above stand for their finite graded copies.

Let $\Gamma := \mathbb{Z}/3\mathbb{Z}$, with $\partial(a) := 1$, $\partial(c) := 1$, and $\partial(b) := 2$. The homogeneous relations have degree zero in Γ . Use semantic types U_i ($i \in \Gamma$). A scored root has type $V_{0,*}$, and a scored nonroot reached by a letter s has type $V_{i,s}$, where i is its degree class. All formal types $V_{i,*}$ are retained in the finite signature. An exact cutoff core may use none of their colors, but the homogeneous construction later gives every type positive capacity and fills unused capacity by diffuse slack colors.

For a weighted record, let

$$\mathcal{U} := \bigsqcup_{i \in \Gamma} C_{U_i}, \quad \mathcal{V} := \bigsqcup_{\substack{i \in \Gamma \\ \tau \in \{*,a,b,c\}}} C_{V_{i,\tau}},$$

and write u and v for the restrictions of the weight vector to $\mathcal{U}$ and $\mathcal{V}$, respectively.

Put

$$\mathcal{A}_{\text{nf}} := \{(\tau, s) : \tau \in \{*, a, c\}, s \in \{a, b, c\}\} \cup \{(b, b)\}.$$

These are exactly the canonical normal-form transitions. Once the first b has been used, only further b -edges are scored. For $(\tau, s) \in \mathcal{A}_{\text{nf}}$ introduce $E_{i,\tau,s} : V_{i,\tau} \rightarrow V_{i+\partial(s),s}$ and the pairing relation $I_{i,\tau} : U_i \rightarrow V_{i,\tau}$. The source relation is $S : Y \rightarrow V_{0,*}$. The channel list itself forces the target of an s -edge to have incoming label s .

All sums involving $E_{i,\tau,s}$ below are restricted to $(\tau, s) \in \mathcal{A}_{\text{nf}}$. For a repaired structure define

$$\begin{aligned} h_{IE}(i, \tau, s) &:= \sum_{I_{i,\tau}(u,v), E_{i,\tau,s}(v,w)} w_u w_w, \\ h_{EE}(i, \tau; s, t) &:= \sum_{E_{i,\tau,s}(v,w), E_{i,\tau,t}(v,w')} w_w w_{w'} \quad (s < t). \end{aligned}$$

Let $\mathcal{G}_E$ be the directed graph on scored colors whose arcs are all repaired E -edges. For each weak scored component $\mathcal{C}$, define

$$B_{\mathcal{C}} \delta_x := \sum_{x \xrightarrow{s} x' \text{ in } \mathcal{C}} r_s \delta_{x'}, \quad C_{\mathcal{C}} := \text{Id} - B_{\mathcal{C}},$$

and let $C := \bigoplus_{\mathcal{C}} C_{\mathcal{C}}$ be the block-diagonal direct sum of the matrices $C_{\mathcal{C}}$. Thus for the scored weight vector v ,

$$(C^{\text{T}} v)_x = v_x - \sum_{x \xrightarrow{s} x'} r_s v_{x'}.$$

When $\mathcal{C}$ is a rooted arborescence, this is exactly the tree matrix from [Section 3.3](#).

Incoming uniqueness makes the ranges of all relations entering one fixed $V_{j,s}$ disjoint. Put

$$\begin{aligned} Z^{\sharp}(v) &:= \sum_{i,\tau} p_2(V_{i,\tau}) + \sum_{i,s} r_s^2 p_2(V_{i,s}) - 2 \sum_{i,\tau,s} r_s h(E_{i,\tau,s}) \\ &\quad + 2 \sum_{i,\tau} \sum_{s < t} r_s r_t h_{EE}(i, \tau; s, t). \end{aligned}$$

If $z = C^{\text{T}} v$ on the underlying scored digraph, then $\|z\|_2^2 \leq Z^{\sharp}(v)$. Indeed, expansion of every parent square gives the displayed edge and sibling terms, while the child-square contribution in the range of an s -edge is bounded by $r_s^2 p_2(V_{j,s})$. Equality holds when every positive nonroot has its intended incoming edge.

Define

$$\begin{aligned} \mathcal{J}^{\sharp} &:= 2h(S) - 2 \sum_i p_2(U_i) - Z^{\sharp}(v) + 2 \sum_{i,\tau} h(I_{i,\tau}) \\ &\quad - 2 \sum_{i,\tau,s} r_s h_{IE}(i, \tau, s). \end{aligned}$$

Let I_0 be the repaired union of the pairing relations. For a scored vertex x , put $z_x := w_x - \sum_{x \xrightarrow{s} x'} r_s w_{x'}$. Because every pairing and every scored-child relation is a partial matching,

$$2 \sum_{i,\tau} h(I_{i,\tau}) - 2 \sum_{i,\tau,s} r_s h_{IE}(i,\tau,s) = 2 \sum_{(u,x) \in I_0} w_u z_x.$$

Moreover, the preceding expansion gives $\sum_{x \in \mathcal{V}} z_x^2 \leq Z^\sharp(v)$. For $x \in \text{ran } I_0$, let u_x be the weight of its unique semantic partner and define

$$\mathcal{J}_{I_0}^S(u, v) := 2h(S) - 2 \sum_{x \in \mathcal{U}} w_x^2 - \sum_{x \in \mathcal{V}} z_x^2 + 2 \sum_{x \in \text{ran } I_0} u_x z_x, \quad z = C^\top v. \quad (4.4)$$

Consequently,

$$\mathcal{J}^\sharp \leq 2h(S) - 2 \sum_{x \in \mathcal{U}} w_x^2 - \sum_{x \in \mathcal{V}} z_x^2 + 2 \sum_{(u,x) \in I_0} w_u z_x = \mathcal{J}_{I_0}^S(u, v). \quad (4.5)$$

Equality holds in the intended finite normal-form trees, because every positive nonroot then has its prescribed incoming edge.

The semantic switch is split by phase. For every $i \in \Gamma$ and $s \in \{a, b, c\}$, put $j := i + \partial(s)$ and use $A_{i,s}: U_i \rightarrow O_{i,s}$ and $K_{i,s}: U_j \rightarrow D_{i,s}$, together with the equality matching

$$T_{i,s}: O_{i,s} \dot{\cup} D_{i,s} \longrightarrow U_j \dot{\cup} Z_{i,s} \dot{\cup} P_{i,s}.$$

The masses satisfy $a_{O_{i,s}} = r_s a_{U_i}$, $a_{D_{i,s}} = a_{U_j}$, and $a_{Z_{i,s}} + a_{P_{i,s}} = r_s a_{U_i}$. Call these 27 aggregate objects the *bridge packages*. For each bridge package $\mathfrak{B}$ of type $A_{i,s}$, $K_{i,s}$, or $T_{i,s}$, put $r_{\mathfrak{B}} := r_s$, 1, or 1, respectively.

When the completed matching defects vanish, Lemma 4.3 makes $A_{i,s}$, $K_{i,s}$, and $T_{i,s}$ total on positive support with the prescribed ratios. The queries below then make their induced maps a total-to-zero model satisfying the two defining relations, the *acc* ideal relation, and nonzero injectivity. Conversely, a ratio-compatible finite cutoff model yields zero defect phase by phase. For each source u , create an $O_{i,s}$ -color of weight $r_s w_u$; for each target $z \in U_j$, create a $D_{i,s}$ -color of weight w_z . Send the O -color to $R_s(u)$ when this image is nonzero and otherwise to $Z_{i,s}$. Send the D -color associated with z to z when $z \notin \text{ran } R_s$ and otherwise to its $P_{i,s}$ -color. Nonzero injectivity makes $T_{i,s}$ a matching, and the balance equations give the required total masses.

4.1.3 Reserves, complements, and the finite query list

We now specify the component signature of every bridge package. An aggregate package $\mathfrak{B}: \mathcal{X}(\mathfrak{B}) \rightarrow \mathcal{Y}(\mathfrak{B})$ consists of pairwise disjoint atomic domain types $(X_\alpha)_{\alpha \in I_{\mathfrak{B}}}$, pairwise disjoint atomic codomain types $(Y_\beta)_{\beta \in J_{\mathfrak{B}}}$, and primary component relations $\mathfrak{B}_{\alpha\beta}: X_\alpha \longrightarrow Y_\beta$ for $(\alpha, \beta) \in E_{\mathfrak{B}}$. An atom $\mathfrak{B}(x, y)$ is shorthand for the unique component determined by the prescribed atomic types of x and y , and

$$a_{\mathcal{X}(\mathfrak{B})} := \sum_{\alpha \in I_{\mathfrak{B}}} a_{X_\alpha}, \quad p_2(\mathcal{X}(\mathfrak{B})) := \sum_{\alpha \in I_{\mathfrak{B}}} p_2(X_\alpha), \quad h(\mathfrak{B}) := \sum_{(\alpha, \beta) \in E_{\mathfrak{B}}} h(\mathfrak{B}_{\alpha\beta}),$$

with analogous notation on the codomain. The complete list is

package	domain components	codomain components	allowed primary components
$A_{i,s}$	U_i	$O_{i,s}$	$U_i \rightarrow O_{i,s}$
$K_{i,s}$	U_j	$D_{i,s}$	$U_j \rightarrow D_{i,s}$
$T_{i,s}$	$O_{i,s}, D_{i,s}$	$U_j, Z_{i,s}, P_{i,s}$	$O \rightarrow U, O \rightarrow Z, D \rightarrow U, D \rightarrow P$

where $j := i + \partial(s)$. We continue to suppress the component superscripts in path formulas.

Let

$$\mathcal{X}_{\text{sq}} := \{U_i : i \in \Gamma\} \cup \{V_{i,\tau} : i \in \Gamma, \tau \in \{*, a, b, c\}\}.$$

For every $X \in \mathcal{X}_{\text{sq}}$, introduce a directed loopless square reserve $Q_X^{\text{sq}} \subseteq X \times X$, whose complete ordered density is $q_X^{\text{sq}} := a_X^2 - p_2(X)$. For every aggregate package $\mathfrak{B}$ and every ordered pair of domain components $\alpha, \alpha' \in I_{\mathfrak{B}}$, introduce a reserve component $N_{\alpha\alpha'}^{\mathfrak{B}, \text{dom}} \subseteq X_\alpha \times X_{\alpha'}$, and similarly introduce $N_{\beta\beta'}^{\mathfrak{B}, \text{cod}} \subseteq Y_\beta \times Y_{\beta'}$

for all $\beta, \beta' \in J_{\mathfrak{B}}$. A same-component reserve is loopless; a cross-component reserve has disjoint endpoint types. Define

$$q_{\mathfrak{B}}(\mathcal{X}(\mathfrak{B})) := \sum_{\alpha, \alpha' \in I_{\mathfrak{B}}} h(N_{\alpha\alpha'}^{\mathfrak{B}, \text{dom}}),$$

$$q_{\mathfrak{B}}(\mathcal{Y}(\mathfrak{B})) := \sum_{\beta, \beta' \in J_{\mathfrak{B}}} h(N_{\beta\beta'}^{\mathfrak{B}, \text{cod}}).$$

When all these reserve components are complete on ordered pairs of distinct positive colors, their values are

$$q_{\mathfrak{B}}(\mathcal{X}(\mathfrak{B})) = a_{\mathcal{X}(\mathfrak{B})}^2 - p_2(\mathcal{X}(\mathfrak{B})), \quad q_{\mathfrak{B}}(\mathcal{Y}(\mathfrak{B})) = a_{\mathcal{Y}(\mathfrak{B})}^2 - p_2(\mathcal{Y}(\mathfrak{B})).$$

For an atomic type X , let $\mathcal{N}_X$ be the finite set consisting of Q_X^{sq} when it exists and every same-component package reserve whose two endpoint types are X . Thus every same-type conflict in the manifest is certified by each positive reserve symbol in $\mathcal{N}_X$; conflicts in the auxiliary O, D, Z, P types are certified by the corresponding package reserve. No zero-coefficient reserve is introduced.

Set

$$\widehat{\Delta}_{r_{\mathfrak{B}}}(\mathfrak{B}) := \frac{1}{2}(r_{\mathfrak{B}}^2 a_{\mathcal{X}(\mathfrak{B})}^2 + a_{\mathcal{Y}(\mathfrak{B})}^2) - r_{\mathfrak{B}} h(\mathfrak{B}) - \frac{1}{2} r_{\mathfrak{B}}^2 q_{\mathfrak{B}}(\mathcal{X}(\mathfrak{B})) - \frac{1}{2} q_{\mathfrak{B}}(\mathcal{Y}(\mathfrak{B})),$$

and let Δ_{br} be the sum over all bridge packages $\mathfrak{B}$. Once the channel-expanded primary relation is a partial matching and the reserves are complete, this is exactly the square defect in [Lemma 4.2](#) on the two disjoint unions.

For every I - E path introduce $C_{i,\tau,s}^{IE} \subseteq U_i \times V_{i+\partial(s),s}$, and for every sibling pair $s < t$ introduce $C_{i,\tau;s,t}^{EE} \subseteq V_{i+\partial(s),s} \times V_{i+\partial(t),t}$. The finite query list contains

$$I_{i,\tau}(u, v), \quad E_{i,\tau,s}(v, w), \quad C_{i,\tau,s}^{IE}(u, w),$$

$$E_{i,\tau,s}(v, w), \quad E_{i,\tau,t}(v, w'), \quad C_{i,\tau;s,t}^{EE}(w, w').$$

After the support repair of [Proposition 4.7](#) and complement completion,

$$h(C_{i,\tau,s}^{IE}) = a_{U_i} a_{V_{i+\partial(s),s}} - h_{IE}(i, \tau, s), \quad (4.6)$$

$$h(C_{i,\tau;s,t}^{EE}) = a_{V_{i+\partial(s),s}} a_{V_{i+\partial(t),t}} - h_{EE}(i, \tau; s, t). \quad (4.7)$$

We make the path queries precise. In the phase-split language used in [Section 4.1](#), let $w := s_1 \cdots s_m$ be one of ba, ab, bc, aca, acc , and put $i_j := i + \sum_{h=1}^j \partial(s_h)$ for $0 \leq j \leq m$, where an empty sum is zero and all indices lie in Γ . The conjunction $\text{NZ}_{i,w}(x_0, x_m)$ consists of fresh variables $x_j \in U_{i_j}$ and $o_j \in O_{i_{j-1}, s_j}$ and the atoms

$$A_{i_{j-1}, s_j}(x_{j-1}, o_j), \quad T_{i_{j-1}, s_j}(o_j, x_j) \quad (1 \leq j \leq m).$$

For $1 \leq q \leq m$, the conjunction $Z_{i,w,q}(x_0)$ uses the same nonzero atoms for $j < q$ and then the atoms

$$A_{i_{q-1}, s_q}(x_{q-1}, o_q), \quad T_{i_{q-1}, s_q}(o_q, z_q), \quad z_q \in Z_{i_{q-1}, s_q},$$

so that the q th letter is the first explicit zero choice. These are finite positive conjunctive templates; the notation introduces no new predicate.

For each defining pair $(w, w') \in \{(ba, ab), (bc, aca)\}$ and each starting phase, include the following expansions.

- (L1) $Z_{i,w,q}(x)$ together with $\text{NZ}_{i,w'}(x, z)$, and the symmetric query, for every possible first-zero position;
- (L2) $\text{NZ}_{i,w}(x, z)$ and $\text{NZ}_{i,w'}(x, z')$ together with either orientation of a positive reserve on the common endpoint type.

Include also $\text{NZ}_{i,acc}(x, z)$, and, for nonzero injectivity, two one-letter nonzero templates with a common image together with a reserve on their two sources. Thus whenever the relevant paths are fully represented on repaired positive support, these queries force the two relator words to vanish together or to have the same nonzero endpoint, force a fully represented acc path to be zero, and force each fully represented nonzero generator action to be injective. Missing bridges are not inferred away here; they are charged by the stopped-tree argument in [Section 4.2](#).

For an aggregate package $\mathfrak{B}$, functionality is expanded as follows over every pair of primary components with the same domain component.

$$\mathfrak{B}_{\alpha\beta}(x, y), \quad \mathfrak{B}_{\alpha\beta'}(x, y'), \quad (\alpha, \beta), (\alpha, \beta') \in E_{\mathfrak{B}},$$

Injectivity is expanded as follows over every pair with the same codomain component.

$$\mathfrak{B}_{\alpha\beta}(x, y), \quad \mathfrak{B}_{\alpha'\beta}(x', y), \quad (\alpha, \beta), (\alpha', \beta) \in E_{\mathfrak{B}}.$$

If the competing variables have distinct atomic types, the two primary atoms alone form a forbidden query. If they have the same atomic type X , then, for every $N \in \mathcal{N}_X$, both queries obtained by adjoining $N(z, z')$ and $N(z', z)$ are included, where z, z' are the competing variables. Thus the four components of each T -package are cross-expanded and their union is forced to be a matching on the disjoint domain and codomain unions.

The same rule is applied to the nonpackage unions. Pairing functionality and injectivity are cross-expanded over all $I_{i,\tau}$ with the same U_i domain or the same scored target type. Incoming uniqueness of E is cross-expanded over every scored-child symbol with the same target type. Outgoing functionality of E is expanded only among symbols with the same parent type and the same label, so distinct labeled children remain allowed. Source functionality is expanded over all S -edges with a common source.

The remaining nonlocal templates are, with $j := i + \partial(s)$,

$$\begin{aligned} &S(y, o), \quad S(y, o'), \quad Q_{V_{0,*}}^{\text{sq}}(o, o'), \\ &I_{i,\tau}(u, v), \quad E_{i,\tau,s}(v, w), \quad A_{i,s}(u, o), \quad T_{i,s}(o, z_0), \quad z_0 \in Z_{i,s}. \end{aligned}$$

For co-reference, for every $N \in \mathcal{N}_{U_j}$ we include both positive queries obtained from the five nonreserve atoms

$$I_{i,\tau}(u, v), \quad E_{i,\tau,s}(v, w), \quad I_{j,s}(u', w), \quad A_{i,s}(u, o), \quad T_{i,s}(o, z) \quad (4.8)$$

by adjoining $N(z, u')$ and $N(u', z)$, respectively. Thus every positive same-component reserve on U_j , not only $Q_{U_j}^{\text{sq}}$, certifies a wrong nonzero co-reference. After reserve completion, the absence of all these queries forces $z = u'$. The first line above enforces source left-functionality, and the second is the zero-edge query. No scored edge can enter $V_{0,*}$ in the graded language, so the sourced-root incoming condition is automatic from the type channels.

To make the expansion canonical, fix the deterministic family orders on the atomic types, relation symbols, reserve symbols, and words that are recorded by the ancillary generator and signature. In every functionality or injectivity group, expand over unordered pairs of relation symbols with repetition; do not add the same pair again in the reverse order. If the competing endpoint types agree, retain both directed orientations of each reserve symbol in $\mathcal{N}_X$. All other indices are expanded in their corresponding fixed order. This convention is used by the ancillary manifest generator and removes every indexing ambiguity.

Together, these templates form the finite query list. All phase choices, channel choices, reserve symbols, and required reserve orientations are expanded, and all path-interior variables are chosen fresh. The next lemma records the resulting finite syntactic checks.

Lemma 4.4. *The expansion rules above produce a finite list of positive typed queries. Every expanded query has a simple binary incidence graph, and every nonreserve atom joins two distinct atomic types. At most one atom of an expanded query has equal endpoint types. When such an atom occurs, it is the reserve that certifies the corresponding conflict.*

Proof. The phase, channel, word, path-position, reserve, and orientation sets are all finite, so the expansion is finite. The complete schema-by-schema manifest is recorded in Table 2; its rows are exactly the outgoing, incoming, source, relator, ideal, nonzero injectivity, zero-edge, co-reference, and complement templates defined above. Interior path variables are fresh in every row.

The grading separates the endpoint types of every nonreserve relation. In a path template, consecutive atoms use alternating fresh variables, and the two relator paths have disjoint interior variables. Thus two nonreserve atoms never use the same unordered pair of formal variables. The table adds at most one reserve or complement atom, and its endpoint pair is different from every nonreserve pair in that row. These incidence checks prove simplicity.

The only equal-type endpoint pairs are the competing endpoints in a functionality, injectivity, source, unequal-endpoint, nonzero injectivity, or co-reference conflict. The expansion rule inserts exactly one chosen reserve on that pair. Complement atoms join distinct graded endpoint types. Consequently every expanded query has at most one same-type atom, as claimed. $\square$

Definition 4.5. Let $\mathcal{R}_{\text{rel}} := \{(ba, ab), (ab, ba), (bc, aca), (aca, bc)\}$. The set $\mathcal{L}_{\text{cell}}$ consists of the following two families of schemas, with all phase choices taken modulo three.

- (i) for $i \in \Gamma$ and $(\tau, s) \in \mathcal{A}_{\text{nf}}$, the old certificate is $I_{i,\tau}(x, v) \wedge E_{i,\tau,s}(v, w)$, and the required path is the one-letter $A_{i,s}T_{i,s}$ path from x to $U_{i+\partial(s)}$;
- (ii) for $i \in \Gamma$ and $(w^-, w^+) \in \mathcal{R}_{\text{rel}}$, the old certificate is $\text{NZ}_{i,w^-}(x, z)$, and the required path is $\text{NZ}_{i,w^+}(x, z')$.

Thus $|\mathcal{L}_{\text{cell}}| = 3|\mathcal{A}_{\text{nf}}| + 3|\mathcal{R}_{\text{rel}}| = 3 \cdot 10 + 3 \cdot 4 = 42$. A concrete local cell in a query-free repaired structure consists of a schema, a base state x , the displayed old certificate, and an incomplete required path together with its maximal fully realized nonzero prefix. Its first missing bridge is either the absent A -edge of the next generator step or, when that A -edge is present, the absent nonzero T -edge to the next U -type. A present T -edge to a zero type is not a missing bridge. It creates a zero-edge query in case (i) and a zero/nonzero mismatch query in case (ii). Hence every incomplete required path in a query-free repaired structure has one of the two stated first-missing forms. Every required word has length at most three.

Proposition 4.6. Under the canonical expansion convention, the base signature has 52 atomic types and 322 positive relation symbols, and the query list $\mathcal{Q}$ has exactly 2456 syntactic instances. Every instance has at most 11 atoms and at most 11 variables. Moreover, $|\mathcal{L}_{\text{cell}}| = 42$.

Proof. The $1 + 3 + 12 + 4 \cdot 9 = 52$ atomic types are Y , the three U -types, the twelve V -types, and the four auxiliary switch families. The positive relation symbols number $1 + 12 + 30 + 15 + 30 + 27 + 54 + 153 = 322$, comprising source, pairing, scored, square-reserve, I - E complement, sibling-complement, primary bridge, and package-reserve symbols, respectively. The query counts by schema are listed in Table 1; their sum is 2456. The two longest queries compare the words bc and aca and add one reserve, giving $4 + 6 + 1 = 11$ atoms and 11 variables. The local-schema count is given in Definition 4.5, and the incidence assertions are those of Lemma 4.4. $\square$

Proposition 4.7. Fix all aggregate atomic type masses. Let Ψ be a rational polynomial in those masses plus a positive rational linear combination of the densities of the relations in the manifest. Every query-free weighted typed structure can be modified without decreasing Ψ to satisfy the following conditions.

- (i) S is left-functional; the union of the pairing relations is a matching on positive support; every A - and K -relation and the channel-expanded union of each T -package is a matching; every scored vertex has at most one incoming E -edge, and for every parent and every fixed label there is at most one outgoing E -edge of that label. Distinct labeled children are allowed.
- (ii) Whenever both sides of a defining relation are fully realized and nonzero, their endpoints agree. A fully realized first-zero side cannot coexist with a fully realized nonzero opposite side; a fully realized acc path is zero; every fully realized nonzero generator action is injective on its nonzero fibers. Whenever a scored child and its semantic action path are fully realized and nonzero, their co-reference is exact.
- (iii) All same-type and package reserves are complete on the repaired positive support.
- (iv) The two complement relations are exactly the complements in (4.6)–(4.7).

The modified structure is in the repair normal form used below.

Proof. For an atomic type X , define a simple graph G_X on C_X as follows. An unordered pair $\{x, x'\}$ is an edge when the nonreserve atoms of one of the same-type conflict schemas in the manifest are satisfied with x, x' in the two competing positions. The graph G_X depends only on the fixed nonreserve relation incidences, not on the weights. During the polarization below we alter only weights, so every G_X remains unchanged until the final deletion of zero-weight colors. By the expansion rule, for every $N \in \mathcal{N}_X$ the

manifest contains both orientations of the query obtained by adjoining N to those nonreserve atoms. Query-freeness therefore implies

$$(x, x'), (x', x) \notin N \quad \text{for every } N \in \mathcal{N}_X$$

whenever $\{x, x'\} \in E(G_X)$.

Suppose that x, x' form an edge of G_X and both have positive weight. Keep $w_x + w_{x'}$ fixed and transfer weight between them. Every nonreserve relation has distinct endpoint types, so its density is affine in the transfer parameter. Every cross-type package reserve is also affine. For a same-type reserve, the only possible quadratic term is the pair $w_x w_{x'}$, and both orientations of that pair are absent by the preceding paragraph. Hence its density is affine as well. The aggregate type masses, and therefore the mass polynomial, remain fixed. The objective is affine on the transfer interval, so one endpoint is no worse than the initial point. Moving to that endpoint decreases the number of positive colors. Iteration terminates with positive support independent in every G_X . Delete all zero-weight colors and their incident relation edges; this changes neither the type masses, the objective, nor query-freeness.

The manifest now gives the assertions in (i) and (ii). Conflicts between different atomic types would themselves be forbidden queries, and same-type conflicts cannot remain on the independent positive support. In particular, outgoing functionality for E is imposed label by label, while incoming uniqueness is imposed across all E -symbols with the same target type. Relator, zero, ideal, nonzero injectivity, and co-reference conclusions are used only when all atoms of the corresponding positive path template are present.

Complete every cross-type package reserve on all ordered pairs of positive colors of its endpoint types. Complete every same-type reserve on all ordered pairs of distinct positive colors. Neither completion creates a query. A query containing an inserted same-type reserve would have its remaining atoms certify an edge of some G_X , but the positive support is independent in G_X ; cross-type reserves do not occur as conflict certificates. The operation weakly increases Ψ and makes the displayed formulas for every $q_{\mathfrak{B}}$ exact. Finally, insert an I - E complement edge precisely when its endpoint pair is not generated by an I - E path, and insert a sibling complement edge precisely when its pair is not generated by two scored children of one parent. Each complement query contains exactly one complement atom, so this completion creates no query and yields (4.6)–(4.7). $\square$

4.2 A stopped-tree estimate

We now prove the uniform upper bound used in Theorem 4.16. The argument has two ingredients. A Bellman potential bounds every part of a scored tree that follows the legal normal-form tree. When a branch first ceases to be legal, a short bridge failure or a weighted ideal descent pays for the stopped cone. This avoids a separate skeleton-embedding argument.

Recall the repaired pairing I_0 , the semantic and scored color sets $\mathcal{U}, \mathcal{V}$, and the functional $\mathcal{J}_{I_0}^S$ from (4.4). Call a weak component of $\mathcal{G}_E$ *sourced* if it contains the target of an S -edge. For a sourced component $\mathcal{C}_j$, let o_j be its source target and put $y_j := \sum_{S(y, o_j)} w_y$ and $C_j := C_{\mathcal{C}_j}$. Let $P_j := V(\mathcal{C}_j) \cap \text{ran } I_0$. Define

$$\mathcal{J}_j := 2y_j v_{o_j} - \sum_{x \in V(\mathcal{C}_j)} z_x^2 - 2 \sum_{x \in P_j} u_x^2 + 2 \sum_{x \in P_j} u_x z_x.$$

For an unsourced component $\mathcal{C}$, define $P_{\mathcal{C}}$ similarly and put

$$\mathcal{J}_{\mathcal{C}} := - \sum_{x \in V(\mathcal{C})} z_x^2 - 2 \sum_{x \in P_{\mathcal{C}}} u_x^2 + 2 \sum_{x \in P_{\mathcal{C}}} u_x z_x.$$

Lemma 4.8. *Every sourced weak scored component is a finite directed arborescence rooted at its unique source target. Moreover,*

$$\mathcal{J}_{I_0}^S(u, v) = \sum_{j \text{ sourced}} \mathcal{J}_j + \sum_{\mathcal{C} \text{ unsourced}} \mathcal{J}_{\mathcal{C}} - 2 \sum_{x \in \mathcal{U} \setminus \text{dom } I_0} w_x^2. \quad (4.9)$$

If y is the total mass of the source type, then

$$\sum_j y_j \leq y, \quad \sum_j y_j^2 \leq y^2. \quad (4.10)$$

Proof. Every scored vertex has indegree at most one. A source target has type $V_{0,*}$ and hence indegree zero. If its weak component has n vertices and m edges, then weak connectivity gives $m \geq n - 1$, while the indegree bound gives $m \leq n - 1$. Hence $m = n - 1$ and the underlying undirected graph is a tree. The indegrees sum to $n - 1$; since the source target has indegree zero, it is the unique vertex of indegree zero and every other vertex has indegree one. Thus the component is an arborescence rooted at its unique source target.

The repaired pairing is a matching, so each paired semantic color belongs to one scored component. Also, $h(S) = \sum_j v_{o_j} \sum_{S(y, o_j)} w_y = \sum_j y_j v_{o_j}$. Splitting the remaining terms by weak component proves (4.9). Source left-functionality gives $\sum_j y_j \leq y$, and nonnegativity gives the second inequality in (4.10). $\square$

For a legal normal word p , define its *continuation potential* by

$$\mathbf{g}(p) := \sum_{q: q \text{ is a legal canonical continuation from } p} r(q)^2.$$

The empty continuation is included. Thus $\mathbf{g}(\emptyset) = \gamma$, and

$$\mathbf{g}(p) = 1 + \sum_{s: ps \text{ is legal}} r_s^2 \mathbf{g}(ps). \quad (4.11)$$

Moreover, $1 \leq \mathbf{g}(p) \leq (1 - r_2)^{-1}$.

A *stopping frontier* in a sourced component is an antichain $B^{\text{pair}} \sqcup B^{\text{unp}}$. Let H consist of the vertices having no proper ancestor in the frontier. We require every vertex of H to have a legal root word, every vertex of $H \setminus B^{\text{unp}}$ to be paired, and every vertex in B^{unp} to be unpaired. No condition is imposed below the frontier.

Lemma 4.9. *Let $(\mathcal{C}_j)_j$ be the sourced components, with source masses y_j . Choose a stopping frontier $B_j^{\text{pair}} \sqcup B_j^{\text{unp}}$ in each component. Then*

$$\sum_j \mathcal{J}_j - \sum_j \sum_{b \in B_j^{\text{pair}}} u_b^2 \leq 2\gamma \sum_j y_j^2 \leq 2\gamma y^2.$$

Proof. Fix a sourced component and write $a_x := y_j r(p_x)$, where p_x is the word on the root path to x . After restricting v and z to $V(\mathcal{C}_j)$, we have $z = C_j^T v$ and $C_j^{-1} \delta_{o_j} = (r(p_x))_x$, so completing squares coordinatewise gives

$$\mathcal{J}_j = \sum_{x \in P_j} \left(2a_x^2 - \frac{1}{2}(z_x - 2a_x)^2 - 2 \left(u_x - \frac{1}{2}z_x \right)^2 \right) + \sum_{x \in V(\mathcal{C}_j) \setminus P_j} (a_x^2 - (z_x - a_x)^2).$$

In particular, a paired coordinate contributes at most $2a_x^2$, and an unpaired coordinate contributes at most a_x^2 .

We first bound a cone stopped at a frontier vertex b . If b is paired, its root contribution after subtracting u_b^2 satisfies

$$2a_b z_b - z_b^2 - 3u_b^2 + 2u_b z_b \leq 2a_b z_b - \frac{2}{3}z_b^2 \leq \frac{3}{2}a_b^2.$$

Every proper descendant contributes at most twice its squared path weight. The total relative squared path weight of all proper descendants is at most $r_2/(1 - r_2)$. Hence the stopped cone, after subtracting u_b^2 , is less than $a_b^2(3/2 + 2r_2/(1 - r_2)) < 2a_b^2$. If b is unpaired, the same argument gives the upper bound $a_b^2(1 + 2r_2/(1 - r_2)) < 2a_b^2$. Since $\mathbf{g}(p_b) \geq 1$, either stopped cone is bounded by $2a_b^2 \mathbf{g}(p_b)$.

Now work upward through the finite tree H . At an internal vertex x with legal word p , the coordinate at the vertex is at most $2a_x^2$. There is at most one child of each label, and every child retained in H has a legal word. By induction and (4.11), the retained subtree contributes at most

$$2a_x^2 + \sum_{s: ps \text{ retained}} 2a_x^2 r_s^2 \mathbf{g}(ps) \leq 2a_x^2 \mathbf{g}(p).$$

At the root this is $2y_j^2 \mathbf{g}(\emptyset) = 2\gamma y_j^2$. Summing over the components and using (4.10) proves the lemma. $\square$

Lemma 4.10. *Every unsourced component $\mathcal{C}$ satisfies $\mathcal{J}_{\mathcal{C}} \leq 0$.*

Proof. A paired coordinate contributes $-2u^2 - z^2 + 2uz = -2(u - z/2)^2 - z^2/2 \leq 0$, and an unpaired coordinate contributes $-z^2 \leq 0$. $\square$

We next control the semantic failures that create the stopping frontier. For a semantic or switch color x , write $\omega(x) := w_x$. For a realized nonzero transition $x \xrightarrow{A_s} o \xrightarrow{T_s} x'$, put

$$\delta(x, s) := \frac{1}{2}(r_s \omega(x) - \omega(o))^2 + \frac{1}{2}(\omega(o) - \omega(x'))^2.$$

Then

$$|\omega(x') - r_s \omega(x)|^2 \leq 4\delta(x, s). \quad (4.12)$$

If the A_s -edge is absent, its package contributes $r_s^2 \omega(x)^2/2$. If the A_s -edge is present but the required nonzero T_s -edge is absent, then

$$\frac{1}{2}(r_s \omega(x) - \omega(o))^2 + \frac{1}{2}\omega(o)^2 \geq \frac{1}{4}r_s^2 \omega(x)^2. \quad (4.13)$$

For a realized path $x_0 \xrightarrow{s_1} \dots \xrightarrow{s_m} x_m$, repeated substitution gives

$$\omega(x_m) = r(s_1 \dots s_m) \omega(x_0) + \sum_{j=1}^m r(s_{j+1} \dots s_m) (\omega(x_j) - r_{s_j} \omega(x_{j-1})). \quad (4.14)$$

The concrete local cells and their first missing bridges are those of [Definition 4.5](#).

Lemma 4.11. *Let*

$$c_{\text{cell}} := \frac{1}{64,000,000,000,000}, \quad C_{\text{cell}} := 15|\mathcal{L}_{\text{cell}}| = 630.$$

Every concrete local cell based at x can be assigned atomic bridge charges of total δ_f such that

$$\delta_f \geq c_{\text{cell}} \omega(x)^2. \quad (4.15)$$

Over all concrete cells in a finite repaired structure,

$$\sum_f \delta_f \leq C_{\text{cell}} \Delta_{\text{br}}. \quad (4.16)$$

Proof. Let the required word be $s_1 \dots s_m$, where $m \leq 3$, and let the first missing bridge occur in step j . For $0 \leq h < j$, put $W_h := r(s_1 \dots s_h) \omega(x)$, where the prefix is empty when $h = 0$. For $1 \leq h < j$, write $\varepsilon_h := \omega(x_h) - r_{s_h} \omega(x_{h-1})$, and let δ_h be the two matched-edge charges in that transition. If $\sum_{1 \leq h < j} \varepsilon_h^2 \geq W_{j-1}^2/16$, then (4.12) gives $\sum_{1 \leq h < j} \delta_h \geq W_{j-1}^2/64$. Otherwise, (4.14), Cauchy-Schwarz, and $j-1 \leq 2$ give

$$|\omega(x_{j-1}) - W_{j-1}| \leq \sqrt{j-1} \left(\sum_{1 \leq h < j} \varepsilon_h^2 \right)^{1/2} < \frac{1}{2} W_{j-1}.$$

Thus $\omega(x_{j-1}) \geq W_{j-1}/2$. The unmatched-domain charge of a missing A -edge is then at least $r_{s_j}^2 W_{j-1}^2/8$; for a missing T -edge, (4.13) gives at least $r_{s_j}^2 W_{j-1}^2/16$. Since $j \leq 3$, all three alternatives imply (4.15).

For the overlap bound, regard an atomic charge as a particular summand in the square decomposition of one bridge package, indexed by its matched edge or its unmatched domain color. Fix such a charge, a schema, and a role in the required path. The charged data determine the semantic state immediately before the corresponding generator step. For a matched A -edge this state is its domain. For a matched T -edge, the edge determines its switch-color source, and the A -matching recovers the preceding semantic state. An unmatched-domain A -charge directly determines that state. In the missing- T case, the matched A -charge again determines it directly, while the unmatched-domain T -charge determines the switch color and the A -matching recovers its unique predecessor.

The part of the required path preceding the charged step is fully realized and nonzero. Its labels are fixed by the schema and the chosen role, so injectivity of the fully realized nonzero generator actions recovers the

base state x by reversing this word. Once x is fixed, the old certificate is unique. In schema (i), the pairing matching determines v , and outgoing E -functionality for the fixed label determines w . In schema (ii), functionality of the realized semantic action determines the old path and its endpoint. Functionality also determines the maximal realized prefix of the required path. Hence a fixed atomic charge occurs in at most one concrete cell with the prescribed schema and role.

There are six realized-prefix roles, corresponding to $1 \leq h < j \leq 3$ and the choice of the matched A - or T -charge, three unmatched- A roles at the first missing step, and six missing- T roles, according to the step and the choice of its matched- A or unmatched- T charge. Summing these fifteen roles over the 42 schemas proves (4.16). $\square$

The only unbounded obstruction is the first entry into the Rees ideal. The following elementary lemma fixes the replacement order used in that descent.

Lemma 4.12. *Let p be a legal normal-form word and suppose that a canonical child ps is the first word of its branch lying in $\mathfrak{J}$. Then $s = c$, and for some $\ell \geq 0$, $m > \ell$, and a possibly empty legal prefix α , one has $p = \alpha a^m c a^\ell$. Writing $d := m - \ell$, fix the derivation*

$$pc = \alpha a^d (a^\ell c a^\ell) c \implies \alpha a^d b^\ell c c \implies \alpha b^\ell a^{d-1} (acc). \quad (4.17)$$

Here $a^n c a^n \implies b^n c$ is expanded recursively. Replace the central aca by bc , move the new b left one position at a time using $ab \rightarrow ba$, and repeat on the remaining $a^{n-1} c a^{n-1}$. In the second arrow, commute $a^d b^\ell$ to $b^\ell a^d$ using $ab \rightarrow ba$, then expose the terminal acc . This convention turns both arrows into a fixed finite sequence of elementary oriented-relator replacements.

Proof. After the first b , the canonical tree permits only further b -edges. Appending a changes only the terminal exponent, and appending b creates no new c -block. Appending c creates the first descent exactly in the displayed situation $m > \ell$. The stated recursive replacements prove the two arrows. $\square$

The next lemma packages the entire descent into one weighted path comparison.

Lemma 4.13. *Let $\mathcal{P}_{\mathfrak{J}}$ be a family of legal paired prefixes p with the following properties. Every prefix of p , including the empty prefix and p itself, is paired and is correctly co-referenced with its fully realized nonzero semantic path; the final c -transition from the partner of p is fully realized and nonzero; and pc is the first word of that branch in $\mathfrak{J}$. Write u_p for the weight of the unique semantic partner of p . Then*

$$\sum_{p \in \mathcal{P}_{\mathfrak{J}}} u_p^2 \leq C_I \Delta_{\text{br}},$$

where

$$C_I := \frac{2t^4 C_{\text{cell}}}{c_{\text{cell}}(1 - r_2)} + \frac{8L_{\text{row}}L_{\text{col}}}{t^2}, \quad L_{\text{row}} := 1 + \frac{1}{1 - r_{\text{max}}}, \quad L_{\text{col}} := 1 + \frac{1}{1 - r_1}.$$

Proof. For each p , follow the fixed elementary derivation from Lemma 4.12 in the repaired partial action. A relator replacement is *supported* if both sides are fully realized and nonzero. Supported replacements have the same endpoint by the relator queries. If every replacement were supported, (4.17) would produce a fully realized nonzero acc -path, contrary to the ideal query. Hence the descent has a first unsupported cell $f(p)$. A represented zero on its required side would create a zero/nonzero query, so $f(p)$ is one of the short missing-bridge cells controlled by Lemma 4.11.

Immediately before that cell, write the current word as $\alpha_f \lambda q_p$, where λ is the realized old relator side, and let $x_{f(p)}$ be the semantic state reached after α_f . All previous replacements are supported. Consequently the realized path from $x_{f(p)}$ labeled λq_p and the final c -transition from the semantic partner ξ_p of p end at the same state y_p . For a realized transition $e: x \xrightarrow{s} x'$, write $\varepsilon_e := \omega(x') - r_s \omega(x)$. Applying (4.14) to the two paths, and using $r(\lambda) = t^3$, gives

$$tu_p = t^3 r(q_p) \omega(x_{f(p)}) + \eta_p, \quad \eta_p := \sum_e A_{p,e} \varepsilon_e, \quad (4.18)$$

where

$$A_{p,e} := \sum_{j: e_{p,j}=e} r(s_{p,j+1} \cdots s_{p,m_p}) - \mathbf{1}_{\{e=e_p^c\}}.$$

Here $(e_{p,1}, \dots, e_{p,m_p})$ is the path labeled λq_p , and e_p^c is the final c -transition.

The suffix weights in one row sum to at most $(1 - r_{\max})^{-1}$, and the distinguished final c -edge contributes at most one. Hence $\sup_p \sum_e |A_{p,e}| \leq L_{\text{row}}$. For the column bound, call (p, j) an occurrence of a transition e when $e_{p,j} = e$, and associate with it the suffix word $h := s_{p,j+1} \dots s_{p,m_p}$. For fixed e and h , there is at most one such occurrence. Indeed, the target of e followed by the functional fully realized path labeled h determines y_p . Injectivity of the fully realized nonzero c -action then determines ξ_p , and the pairing matching determines the paired scored vertex and hence the prefix p . Once p is fixed, the suffix word determines j , because suffixes arising at distinct positions have distinct lengths. Moreover, e can equal the distinguished final edge e_p^c for at most one p : its source determines ξ_p , after which the pairing matching again determines p . Therefore, for every e ,

$$\sum_p |A_{p,e}| \leq 1 + \sum_{h \in \{a,b,c\}^*} r(h) = 1 + \sum_{\ell \geq 0} r_1^\ell = 1 + \frac{1}{1 - r_1} = L_{\text{col}}.$$

For completeness, the finite-matrix Schur bound says that

$$\|A\|_{2 \rightarrow 2}^2 \leq \left(\sup_p \sum_e |A_{p,e}| \right) \left(\sup_e \sum_p |A_{p,e}| \right).$$

Indeed, if R and C denote the two suprema, then Cauchy–Schwarz gives

$$\sum_p \left| \sum_e A_{p,e} \xi_e \right|^2 \leq R \sum_{p,e} |A_{p,e}| |\xi_e|^2 \leq RC \sum_e |\xi_e|^2.$$

Distinct semantic transitions use disjoint matched A - and T -edges after the repair. Applying this bound and (4.12) therefore gives

$$\sum_p \eta_p^2 \leq L_{\text{row}} L_{\text{col}} \sum_e \varepsilon_e^2 \leq 4L_{\text{row}} L_{\text{col}} \Delta_{\text{br}}. \quad (4.19)$$

For fixed data (f, q) , at most one prefix p occurs. Indeed, the concrete cell f fixes its base state and its fully realized old relator side. Functionality therefore makes the endpoint of that side followed by the realized path labeled q , namely y_p , unique. Since the final c -transition from ξ_p to y_p is fully realized and nonzero, injectivity of the nonzero c -action determines at most one ξ_p . The pairing matching then determines the paired scored vertex and hence p . From (4.18),

$$u_p^2 \leq 2t^4 r(q_p)^2 \omega(x_{f(p)})^2 + \frac{2}{t^2} \eta_p^2.$$

Since $\sum_q r(q)^2 = (1 - r_2)^{-1}$, summing first over all right contexts and then over all failed cells gives

$$\sum_p u_p^2 \leq \frac{2t^4}{1 - r_2} \sum_f \omega(x_f)^2 + \frac{2}{t^2} \sum_p \eta_p^2.$$

The local-cell lemma yields $\sum_f \omega(x_f)^2 \leq (C_{\text{cell}}/c_{\text{cell}}) \Delta_{\text{br}}$. Combining this bound with (4.19) proves the stated value of C_I . $\square$

Theorem 4.14. *Every finite repaired structure satisfies*

$$\mathcal{J}_{I_0}^S(u, v) \leq 2y^2 \gamma + C_B \Delta_{\text{br}}, \quad C_B := \frac{C_{\text{cell}}}{c_{\text{cell}}} + C_I. \quad (4.20)$$

Proof. Unsourced components and semantic colors outside $\text{dom } I_0$ contribute at most zero by Lemmas 4.8 and 4.10. We construct a stopping frontier in every sourced component. If the root is unpaired, stop at the root. Otherwise traverse away from it. At a paired legal vertex q , first inspect the semantic transitions corresponding to all present scored children. If one is not fully realized and nonzero, put q in B_{loc} and stop its whole cone. If all these transitions are fully realized and nonzero but one child is the first word of its branch in $\mathfrak{J}$, put q in $B_{\mathfrak{J}}$ and again stop its whole cone. Otherwise every child word is legal. Stop the cone of each unpaired child at that child and put it in B_{unp} ; for every paired child, co-reference identifies

its semantic partner with the transition endpoint, and the traversal continues. Once a cone is stopped, it is not inspected again. Thus $B_{\text{loc}} \sqcup B_{\mathfrak{J}} \sqcup B_{\text{unp}}$ is an antichain and defines a stopping frontier.

A local boundary supplies a concrete one-letter cell based at its semantic partner. Hence Lemma 4.11 gives $\sum_{b \in B_{\text{loc}}} u_b^2 \leq (C_{\text{cell}}/c_{\text{cell}})\Delta_{\text{br}}$. Every prefix of a vertex in $B_{\mathfrak{J}}$ survived the traversal, so it satisfies the hypotheses of Lemma 4.13. Therefore $\sum_{b \in B_{\mathfrak{J}}} u_b^2 \leq C_I \Delta_{\text{br}}$. Consequently,

$$\sum_{b \in B_{\text{loc}} \cup B_{\mathfrak{J}}} u_b^2 \leq C_B \Delta_{\text{br}}. \quad (4.21)$$

Applying Lemma 4.9 to the resulting frontiers and then using (4.21) proves (4.20). $\square$

4.3 Choosing the positive coefficients

We now choose the coefficients that turn the stopped-tree estimate into an exact finite optimization. Fix the explicit rational choice

$$M_0 := \frac{C_B}{4} + 2 = \frac{70,242,590,880,006,679,213,978}{6,967,089}, \quad (4.22)$$

which is strictly larger than $C_B/4 + 1$, and define $\Phi := \mathcal{J}^\sharp/4 - M_0 \Delta_{\text{br}}$. For $X \in \mathcal{X}_{\text{sq}}$, put $\kappa_{U_i} := 1/2$, $\kappa_{V_{i,*}} := 1/4$, and $\kappa_{V_{i,s}} := (1 + r_s^2)/4$. After the repair and completion of Proposition 4.7, Φ is the mass polynomial plus positive relation densities given in the following table.

relation class	coefficient	mass baseline
S	$1/2$	0
$I_{i,\tau}$	$1/2$	0
$E_{i,\tau,s}$	$r_s/2$	0
$Q_X^{\text{sq}}, X \in \mathcal{X}_{\text{sq}}$	κ_X	$-\kappa_X a_X^2$
$C_{i,\tau,s}^{IE}$	$r_s/2$	$-(r_s/2) a_{U_i} a_{V_{i+\partial(s),s}}$
$C_{i,\tau;s,t}^{EE}$	$r_s r_t/2$	$-(r_s r_t/2) a_{V_{i+\partial(s),s}} a_{V_{i+\partial(t),t}}$
bridge component of $\mathfrak{B}$	$M_0 r_{\mathfrak{B}}$	$-\frac{M_0}{2} (r_{\mathfrak{B}}^2 a_{\mathcal{X}}^2 + a_{\mathfrak{J}}^2)$
ordered domain reserve of $\mathfrak{B}$	$M_0 r_{\mathfrak{B}}^2/2$	included above
ordered codomain reserve of $\mathfrak{B}$	$M_0/2$	included above

The bridge baseline is included once per package, not once per atomic component. Denote the sum of the baselines by $\mathcal{B}(\mathbf{a})$ and the finite positive relation list by $\mathcal{R}$, with coefficients $c_R > 0$. For $R \in \mathcal{R}$, write $R: X_R \rightarrow Y_R$ for its ordered endpoint types. The coefficient identity is

$$\Phi = \mathcal{B}(\mathbf{a}) + \sum_{R \in \mathcal{R}} c_R h(R). \quad (4.23)$$

We call

$$\text{Score}(\mathfrak{R}) := \sum_{R \in \mathcal{R}} c_R h_{\mathfrak{R}}(R)$$

the *positive relation score* of a weighted record $\mathfrak{R}$. The square-reserve rows use $-\kappa_X p_2(X) = -\kappa_X a_X^2 + \kappa_X q_X^{\text{sq}}$. The complement rows use (4.6)–(4.7), and the last three rows are exactly $-M_0 \hat{\Delta}_{r_{\mathfrak{B}}}(\mathfrak{B})$.

By (4.5) and Theorem 4.14,

$$\Phi \leq \frac{1}{2} y^2 \gamma + \left(\frac{C_B}{4} - M_0 \right) \Delta_{\text{br}} \leq \frac{1}{2} y^2 \gamma - \Delta_{\text{br}}.$$

For an intended finite legal tree, every nonroot has its intended incoming edge, $\Delta_{\text{br}} = 0$, and $\max \Phi = y^2 g_T/2$.

Lemma 4.15. *Fix a cutoff tree T_D , a rational source mass $y \geq 0$, and a positive rational capacity vector with source coordinate y . Assume that the vector satisfies the switch-balance equations and strictly contains the exact optimizer and its auxiliary switch masses in every nonsource type. There exists a sequence of finite query-free records with exactly those aggregate type masses. Along this sequence, the bridge defect Δ_{br} is $O(n^{-1})$, and*

$$\Phi \longrightarrow \frac{1}{2} y^2 g_D. \quad (4.24)$$

The implied constant depends only on the fixed capacities and the finite signature.

Proof. Place the exact cutoff optimizer, its pairing and scored edges, and the exact zero-defect switch model inside the prescribed capacities. For each atomic type X , let $\rho_X \geq 0$ be the remaining mass. Add n fresh slack colors of type X , each of weight ρ_X/n ; types with $\rho_X = 0$ need no slack colors. Put no source, pairing, scored-child, or primary switch edge on a slack color. Keep all primary relations on the exact core.

There is no positive conflict pair involving a slack color. Indeed, in every same-type conflict schema each competing variable is incident with at least one nonreserve primary atom, whereas slack colors have no incident source, pairing, scored-child, or primary switch edge. A conflict pair contained entirely in the exact core is also impossible, because the exact cutoff model satisfies all functionality, injectivity, relator, ideal, zero, and co-reference constraints. Consequently the positive support contains no conflict pair at all.

Complete every reserve on every ordered pair of distinct positive colors. Define each I - E and sibling complement to be the exact complement of its path relation. The resulting record is query-free. Indeed, a query containing only primary atoms would lie entirely in the exact core and is absent there. A query containing a reserve has its other atoms certify a conflict pair, but no positive conflict pair exists. A query containing a complement has its endpoint pair generated by the corresponding path, and that complement edge was omitted. These are all rows of the manifest.

After reserve completion, every bridge package equals the square defect in [Lemma 4.2](#). Exact core pairs contribute zero. All slack colors are unmatched, so a package with domain components X and codomain components Y contributes at most

$$\frac{1}{2n} \left(r_{\mathfrak{B}}^2 \sum_X \rho_X^2 + \sum_Y \rho_Y^2 \right).$$

Summing over the fixed finite package list shows that Δ_{br} is $O(n^{-1})$.

Finally, use the identity $\Phi = \mathcal{J}^\sharp/4 - M_0\Delta_{\text{br}}$. The primary relation terms of $\mathcal{J}^\sharp$ are those of the exact core. Every additional square term is a sum of squared slack weights and is $O(n^{-1})$; the completed reserves and complements satisfy the exact coefficient identities. Hence $\mathcal{J}^\sharp$ tends to the exact cutoff value $2y^2g_D$, while the bridge defect tends to zero. These limits prove [\(4.24\)](#). $\square$

Theorem 4.16. *There exist finite data consisting of an atomic type set $\mathcal{T}_0$ with source type Y , a list $\mathcal{R}$ of directed typed binary relations, with every same-type relation loopless, a list $\mathcal{Q}$ of positive typed queries, positive rational coefficients c_R , a rational quadratic polynomial $\mathcal{B}(\mathbf{a})$ in the type masses, and a defect functional Δ with rational coefficients. The defect is nonnegative on repaired structures, and the following statements hold.*

- (i) *Every query-free weighted structure can be repaired, without changing its type masses or decreasing Score. On the repaired record,*

$$\Phi = \mathcal{B}(\mathbf{a}) + \text{Score}.$$

- (ii) *If y is the total mass of the source type, then the mass actually attached by S is at most y , and every repaired structure satisfies*

$$\Phi \leq \frac{1}{2}y^2\gamma - \Delta.$$

- (iii) *For every cutoff tree T_D , every source mass y , and every positive rational capacity vector with source coordinate y , suppose that the vector satisfies the switch balances and strictly contains the cutoff optimizer and its auxiliary masses in every nonsource type. Then there exist finite query-free records with those type masses for which*

$$\Phi \longrightarrow \frac{1}{2}y^2g_D, \quad \Delta \longrightarrow 0.$$

- (iv) *All data and constructions are effective. The base signature has 52 atomic types and 322 positive relation symbols, and the canonical manifest contains 2456 expanded queries. Every expanded query has a simple binary incidence graph and at most one same-type atom.*

Proof of Theorem 4.16. The typed-record convention is Definition 4.1, and the finite signature and query list are those constructed in Section 4.1; finiteness and the incidence assertions are proved in Lemma 4.4. The repair theorem Proposition 4.7 and the coefficient identity (4.23) give part (i). By (4.5) and Theorem 4.14,

$$\frac{1}{4}\mathcal{J}^\# - M_0\Delta_{\text{br}} \leq \frac{1}{2}y^2\gamma + \left(\frac{C_B}{4} - M_0\right)\Delta_{\text{br}}.$$

The explicit choice in (4.22) satisfies $M_0 > C_B/4 + 1$ and gives part (ii), with $\Delta := \Delta_{\text{br}}$. In the repair normal form, Lemma 4.2 expresses this functional as a sum of atomic squares, so it is nonnegative on repaired structures. The cutoff models in Proposition 3.10 and Lemma 4.15 give part (iii). The exact finite counts in part (iv) follow from Proposition 4.6. Every coefficient is obtained by rational arithmetic, and the deterministic expansion and verification procedure is recorded in Sections A and B. $\square$

5 From the finite system to untyped, unordered hypergraphs

The goal of this section is to turn Theorem 4.16 into a finite family of untyped, unordered forbidden hypergraphs. We first give every signal profile exactly the same multiset of atomic types. This homogeneous choice removes the extra concentration and semialgebraic arguments that would arise if only the filler coordinates were balanced. We then encode the typed queries by ordered gadgets, attach the required pins on a common core, and use an outer-ordering construction to remove the chosen vertex order. Finally, we compare arbitrary deficient palettes with the intended ones and obtain the exact density formula.

5.1 Homogeneous data

We begin by choosing all homogeneous data simultaneously. Recall r_1 from (3.9) and put $U_* := (1 - r_1)^{-1}$ and $V_* := 2(1 - r_1)^{-2}$.

Lemma 5.1. *For every finite subtree T of the legal normal-form tree,*

$$\|h_T\|_1 \leq \frac{1}{1 - r_1}, \quad \|C_T^{-\top} h_T\|_1 \leq \frac{1}{(1 - r_{\max})(1 - r_1)} \leq \frac{1}{(1 - r_1)^2}. \quad (5.1)$$

Proof. At distance m from the root, the sum of the path weights is at most $(r_a + r_b + r_c)^m = r_1^m$, because a legal tree is a subtree of the free three-letter tree. Hence

$$\|h_T\|_1 = \sum_{x \in T} r(x) \leq \sum_{m \geq 0} r_1^m = \frac{1}{1 - r_1}.$$

Moreover, $C_T^{-\top} = \sum_{k \geq 0} (B_T^\top)^k$ is a finite sum. Every row of B_T contains at most one nonzero entry, because every nonroot vertex has a unique parent, and that entry is at most $r_{\max}$. Therefore $\|B_T^\top\|_1 = \|B_T\|_\infty \leq r_{\max}$, and

$$\|C_T^{-\top} h_T\|_1 \leq \sum_{k \geq 0} r_{\max}^k \|h_T\|_1 \leq \frac{1}{(1 - r_{\max})(1 - r_1)}.$$

Since $r_{\max} \leq r_1$, the final inequality in (5.1) follows. $\square$

Construction 5.2. *Let $\mathcal{T} := \mathcal{T}_0 \sqcup \{F_1, F_2\}$, where the two filler types are not endpoints of any relation in $\mathcal{R}$. Put $p_Y := [2(1 + 24U_*(2 + r_1) + 24V_*)]^{-1} = 6241/1810514$, and set*

$$\begin{aligned} p_{U_i} &:= 4p_Y U_*, & p_{V_{i,\tau}} &:= 2p_Y V_*, \\ p_{O_{i,s}} &:= r_s p_{U_i}, & p_{D_{i,s}} &:= p_{U_j}, \\ p_{Z_{i,s}} &:= \frac{1}{2} r_s p_{U_i}, & p_{P_{i,s}} &:= \frac{1}{2} r_s p_{U_i} \quad (U_i \xrightarrow{s} U_j), \\ p_{F_1} &:= \frac{1}{4}, & p_{F_2} &:= \frac{1}{4}. \end{aligned} \quad (5.2)$$

The nonfiller mass is $p_Y[1 + 24U_*(2 + r_1) + 24V_*] = 1/2$, so these numbers form a positive rational probability vector $\mathbf{p} = (p_v)_{v \in \mathcal{T}}$ on the 52 endpoint types together with the two private filler types. Write

$\mathbf{p}_0 := \mathbf{p}|_{\tau_0}$. Finally, set

$$\begin{aligned} L_0 &:= 72,271,663,407,397,359, \\ M &:= 8,527,079,511,467,783,033,168,342,073,204, \\ N &:= 3,621,028, \quad K := 4N = 14,484,112. \end{aligned} \tag{5.3}$$

Lemma 5.3. *The probability vector and integers in [Construction 5.2](#) have the following properties.*

- (i) *The source mass p_Y is positive, the switch-balance equations hold, and every finite cutoff optimizer and its auxiliary switch masses fit strictly inside the nonsource capacities.*
- (ii) *For every $R \in \mathcal{R}$, one has $m_R := L_0 c_R p_{X_R} p_{Y_R} \in \mathbb{N}$. Writing $M := \sum_R m_R$ and $d_v := N p_v$, all d_v are positive integers, $d_v \geq 2$ for endpoint types, and $K = 4N$.*
- (iii) *There exist integers s_X , one for each endpoint type X , such that every one of the M signal copies has a distinct type word of length N , every such word contains exactly d_v copies of type v , and a copy for $R: X \rightarrow Y$ uses the active coordinates*

$$j_X(R) := s_Y + \mathbf{1}_{\{s_X < s_Y\}}, \tag{5.4}$$

$$j_Y(R) := s_X + \mathbf{1}_{\{s_Y < s_X\}}. \tag{5.5}$$

when $X \neq Y$. When $X = Y$, it uses the consecutive coordinates s_X and $s_X + 1$ in the order of the atom. Here $2 \leq s_X < s_X + 1 \leq N - 1$, and the two-position sets $\{s_X, s_X + 1\}$ are pairwise disjoint and separated by at least one unused position.

- (iv) *With*

$$\Theta(\mathbf{u}) := \prod_{v \in \mathcal{T}} u_v^{d_v}, \quad \theta_* := \Theta(\mathbf{p}),$$

and

$$\beta(x) := L_0 \left(\frac{1}{2} p_Y^2 x - \mathcal{B}(\mathbf{p}_0) \right) - M, \quad f_x(z) := z(1 + \beta(x)z),$$

we have

$$0 < f_x(\theta_*) < 1, \quad 1 + 2\beta(x)\theta_* > 0 \quad (1 \leq x \leq g_{\text{ub}}). \tag{5.6}$$

Proof. Apply [Lemma 5.1](#) to a cutoff optimizer with source mass p_Y . Then $\|u^*\|_1 \leq p_Y U_*$ and $\|v^*\|_1 \leq p_Y V_*$. The capacities in (5.2) are respectively $4p_Y U_*$ and $2p_Y V_*$ for each semantic and scored type, so these types have strict room. In particular,

$$a_{U_i}^* \leq p_Y U_*, \quad a_{V_{i,\tau}}^* \leq p_Y V_*.$$

The auxiliary masses in the exact switch model satisfy

$$\begin{aligned} a_{O_{i,s}}^* &= r_s a_{U_i}^* < 4r_s p_Y U_* = p_{O_{i,s}}, & a_{D_{i,s}}^* &= a_{U_j}^* < 4p_Y U_* = p_{D_{i,s}}, \\ a_{Z_{i,s}}^*, a_{P_{i,s}}^* &\leq r_s a_{U_i}^* < 2r_s p_Y U_* = p_{Z_{i,s}} = p_{P_{i,s}}. \end{aligned}$$

The displayed assignments to O, D, Z, P also satisfy the switch-balance equations. These bounds and identities prove part (i).

There are 322 positive relation symbols. Direct substitution of their rational coefficients and the masses in (5.2) shows that every product $c_R p_{X_R} p_{Y_R}$ has denominator, in lowest terms, equal to either

$$819,490,236,049 \quad \text{or} \quad 72,271,663,407,397,359.$$

The first divides the second, which is L_0 . Thus all numbers $m_R = L_0 c_R p_{X_R} p_{Y_R}$ are positive integers. Their exact sum is the integer M in (5.3); the complete relation-by-relation certificate is described in [Section B](#).

All nonfiller masses other than p_Y have denominator 905257, while $p_Y = 6241/(2 \cdot 905257)$ and the two filler masses are $1/4$. Since $N = 4 \cdot 905257$, all $d_v = N p_v$ are positive integers. Their values, grouped by type orbit, are

$$\begin{array}{cccccccccc} v & Y & U_i & V_{i,\tau} & O_{i,a/c} & O_{i,b} & D_{i,s} & Z/P_{i,a/c} & Z/P_{i,b} & F_1, F_2 \\ d_v & 12482 & 63200 & 80000 & 6320 & 632 & 63200 & 3160 & 316 & 905257. \end{array} \tag{5.7}$$

In particular, every endpoint type has multiplicity at least 316. These computations prove the numerical assertions in part (ii).

Order the 52 endpoint types as follows: first Y ; then U_i by $i = 0, 1, 2$; then $V_{i,\tau}$ lexicographically by i and $\tau \in (*, a, b, c)$; and finally the four families O, D, Z, P , within each family lexicographically by $i \in (0, 1, 2)$ and $s \in (a, b, c)$. Call this list $X_0, \dots, X_{51}$ and set $s_{X_q} := 2 + 3q$ for $0 \leq q \leq 51$. The resulting slots lie in $\{2, \dots, N-1\}$ and are pairwise separated by one unused position. Fix the active entries of a signal word according to (5.4)–(5.5). Active endpoint types are never fillers. After fixing the two active entries, process the nonfiller types in the displayed order and place all remaining copies of each type in the least currently unassigned coordinates. Exactly $N/2$ positions then remain for the $N/4$ copies of F_1 and the $N/4$ copies of F_2 . Consequently every endpoint specification has at least

$$\binom{N/2}{N/4} \geq 2^{N/4} = 2^{905257} > 2^{103} > M$$

completions. Greedily choosing the lexicographically first unused completion therefore gives a distinct type word for every signal copy. The active-slot rank calculation is verified in Lemma 5.8. The completion count and rank calculation prove part (iii).

It remains to check part (iv), for which no asymptotic choice is needed. Exact rational evaluation gives

$$\mathcal{B}(\mathbf{p}_0) = -\frac{477,982,848,835,180,189,012,746,061,632}{8,030,184,823,044,151}, \quad (5.8)$$

$$\beta(1) = -\frac{33,801,870,975,609,290,652,993,973,554,657}{8},$$

$$\beta(g_{\text{ub}}) = -4,225,233,871,951,161,331,615,439,107,266. \quad (5.9)$$

The function β is increasing, both endpoint values are negative, and $|\beta(x)| < 2^{102}$ on $[1, g_{\text{ub}}]$. Since $d_{F_1} = d_{F_2} = N/4$,

$$0 < \theta_* = \prod_v p_v^{d_v} \leq (1/4)^{N/2} = 2^{-N}.$$

Hence throughout this interval,

$$1 + \beta(x)\theta_* > 1 - 2^{102-N} > 0, \quad 1 + 2\beta(x)\theta_* > 1 - 2^{103-N} > 0.$$

Moreover $\beta(x) < 0$, so $0 < f_x(\theta_*) < \theta_* < 1$. These inequalities prove (5.6) and complete the proof. $\square$

For a finite typed record with color classes $(C_v)_{v \in \mathcal{T}}$ and a type word $\mathbf{t} = (t_1, \dots, t_N)$, write $B(\mathbf{t}) := \prod_{i=1}^N C_{t_i}$. Every nonsignal box $B(\mathbf{t})$ is completely allowed. If $\mathbf{t}$ is a signal copy for $R: X \rightarrow Y$, its allowed tuples are those $q \in B(\mathbf{t})$ satisfying $(q_{j_X(R)}, q_{j_Y(R)}) \in R$, with the ordered consecutive coordinates used when $X = Y$. The *semantic block* is the disjoint union of these allowed type boxes, and the *certification block* is the Cartesian product of the prescribed singleton-type color classes. In a positive half, the certification coordinates are $[N]$ and the semantic coordinates are $N + [N]$. A *base half* is the Cartesian product of its certification and semantic block relations; the full *intended palette* is the Cartesian product of this half with its coordinate reverse.

5.2 Removing types and vertex orders

We next remove the auxiliary types and the chosen vertex order. For each ordered query gadget we first build a canonical palette, then attach the prescribed signal and certification pins on a common core. An outer-ordering design replaces the ordered gadget by an unordered hypergraph while preserving colorability in both directions. Only these finite gadgets, together with the shell stars that enforce the type boxes, enter the forbidden family.

For a finite ordered K -graph $(J, <)$ with at least one edge, its *canonical palette* $\text{Can}(J, <)$ has one color for each covered $(K-1)$ -face and one admissible tuple for each edge, in the order of its facets. If J is edgeless, let $\text{Can}(J, <)$ have one dummy color and no admissible tuples.

Lemma 5.4. *For every finite ordered K -graph $(J, <)$ with $K \geq 3$, there exists an effectively constructible finite K -graph $\mathfrak{D}(J)$ such that, for every K -palette Q ,*

$$\mathfrak{D}(J) \text{ is } Q\text{-colorable} \iff \text{Can}(J, <) \rightarrow Q \text{ or } \text{rev Can}(J, <) \rightarrow Q. \quad (5.10)$$

Proof. If $E(J) = \emptyset$, take $\mathfrak{D}(J)$ to be a finite edgeless K -graph. It is Q -colorable for every Q , while the canonical palette of J has one dummy color and no admissible tuple, and hence admits a homomorphism to every nonempty-color palette Q ; thus (5.10) is immediate. We may therefore assume $E(J) \neq \emptyset$. Delete isolated vertices from J , let $m := |V(J)|$, and put $R_0 := (m - 1)^2 + 1$. For a sufficiently large integer $n_{\text{des}} \geq R_0$, fix the reference order on $[n_{\text{des}}]$. For a second total order $\prec$, Erdős-Szekeres [6] implies that every R_0 -set contains an m -set that is increasing or decreasing between the two orders. Counting pairs consisting of an R_0 -set and one such m -subset gives at least

$$\frac{\binom{n_{\text{des}}}{R_0}}{\binom{n_{\text{des}}-m}{R_0-m}} = \frac{\binom{n_{\text{des}}}{m}}{\binom{R_0}{m}} \geq c_m n_{\text{des}}^m$$

monotone m -sets, where one may fix a positive rational c_m depending only on m .

Retain each m -set independently with probability

$$p_{\text{des}} := \frac{C_m \log n_{\text{des}}}{n_{\text{des}}^{m-1}}, \quad C_m > \frac{16}{c_m}.$$

For a fixed order $\prec$, let $X_{\prec}$ denote the number of retained monotone blocks and let $\lambda_{\prec} := \mathbb{E}X_{\prec}$. Then $\lambda_{\prec} \geq c_m C_m n_{\text{des}} \log n_{\text{des}}$, and the lower-tail Chernoff bound gives

$$\Pr(X_{\prec} < \lambda_{\prec}/2) \leq e^{-\lambda_{\prec}/8}.$$

Since $n_{\text{des}}! \leq e^{n_{\text{des}} \log n_{\text{des}}}$, the union bound shows that the event $\mathcal{A}$ that every order has at least $(c_m C_m/2) n_{\text{des}} \log n_{\text{des}}$ retained monotone blocks has probability tending to one.

Let Y be the number of retained block pairs meeting in at least two vertices. There are $O_m(n_{\text{des}}^{2m-2})$ candidate pairs, and hence

$$\mathbb{E}Y = O_m(n_{\text{des}}^{2m-2} p_{\text{des}}^2) = O_m((\log n_{\text{des}})^2).$$

Markov's inequality gives $\Pr(Y \leq 2\mathbb{E}Y) \geq 1/2$. For sufficiently large n_{des} , this event and $\mathcal{A}$ occur simultaneously with positive probability. Delete one block from each remaining bad pair. At most $O_m((\log n_{\text{des}})^2)$ blocks are deleted, whereas every order began with $\Omega_m(n_{\text{des}} \log n_{\text{des}})$ monotone blocks. Thus every order still has a monotone block and the surviving m -uniform design D is linear.

The existence proof is effective in the literal finite sense. Enumerate $n_{\text{des}} = m, m+1, \dots$, then enumerate all families of m -subsets of $[n_{\text{des}}]$, and accept the first family that is linear and has a monotone block for each of the finitely many total orders. The preceding argument proves that this search terminates.

Place an order-preserving copy of $(J, <)$ on every block of D and let $\mathfrak{D}(J)$ be their union. Distinct blocks meet in at most one vertex, so their covered $(K-1)$ -shadows are disjoint. A homomorphism from $\text{Can}(J, <)$ or its reverse gives a Q -coloring using the reference order or its reverse. Conversely, the order induced by a Q -coloring contains a monotone block of D ; restriction to its copy gives one of the two homomorphisms in (5.10). $\square$

5.3 Certification, extraction, and pinned lifts

One base half has d_v singleton certification coordinates of type v for each $v \in \mathcal{T}$, followed by the semantic block. The full coordinate list consists of the base half followed by its reverse. Let $\mathcal{A}_i \subseteq \mathcal{T}$ be the allowed type set at coordinate i . It is a singleton in a certification coordinate and $\mathcal{T}$ in a semantic coordinate. The *type-box palette* is

$$P_{\text{type}} := \left(\mathcal{T}, \prod_{i=1}^K \mathcal{A}_i \right).$$

The coordinate sets are palindromic, so $\text{rev}(P_{\text{type}}) = P_{\text{type}}$.

Fix a target palette Q , a map from $C(Q)$ to $\mathcal{T}$ that assigns a type to each color, and one of the two halves $\varepsilon \in \{+, -\}$; in the negative half, reverse the coordinate order first. In the statements below we assume either that Q is nondegenerate, or that every admissible record in a selected signal copy has distinct active entries whenever the endpoint types agree. The latter condition holds for every intended palette because every same-type typed relation is loopless. A color c of type v is ε -certified if it occurs in an admissible tuple at a singleton- v certification coordinate of that half. A semantic record in a signal copy with type word $\mathbf{t}$ is *wholly certified* when each of the N coordinates of that selected semantic block

has type t_i and is ε -certified. For $R: X \rightarrow Y$, let H_R^ε be the union, over all signal copies for R , of the ordered active-pair projections of wholly certified admissible records. If $X = Y$, the diagonal is excluded, consistently with looplessness.

Definition 5.5. Let $\mathbf{q}$ be an expanded typed query. Unsatisfiable atoms of the form $R(x, x)$ for a loopless relation are omitted from the manifest, so every atom has two distinct formal endpoint variables. Fix a half ε and, for every atom $\alpha = R(x, y)$, choose one signal copy for R .

For each query variable x introduce one formal color c_x . For each atom α , form the formal K -tuple $T_\alpha := (c_{\alpha,1}, \dots, c_{\alpha,K})$. At the two active coordinates put c_x and c_y in the prescribed order. Every other $c_{\alpha,i}$ is a new private color, distinct from all query colors and from every private color belonging to another coordinate or record. Thus the only prescribed identifications among atom records are those imposed by repeated query variables.

For every coordinate i of the selected semantic block of T_α , choose one singleton certification coordinate $k(\alpha, i)$ of the type prescribed by the selected signal word. Add the pin tuple

$$P_{\alpha,i} := (p_{\alpha,i,1}, \dots, p_{\alpha,i,K}), \quad p_{\alpha,i,k(\alpha,i)} := c_{\alpha,i},$$

taking all other entries to be new and private. Repeated occurrences of one query variable may therefore have several pin tuples sharing the same formal color. The pinned full-record lift L is the K -palette whose color set is the set of all formal colors just defined and whose admissible tuples are precisely the atom tuples T_α and the pin tuples $P_{\alpha,i}$. We enumerate the finitely many choices of signal copies and certification coordinates.

Lemma 5.6. Assume that Q is nondegenerate, or that every admissible record in each selected signal copy whose endpoint types agree has distinct active entries. Then an expanded query $\mathbf{q}$ occurs in the extracted relations $(H_R^\varepsilon)_R$ if and only if one of the pinned full-record lifts from Definition 5.5 admits a palette homomorphism to Q .

Proof. Given an occurrence, choose for each atom a wholly certified admissible full tuple witnessing its active pair. Send each query color c_x to the color assigned to x by the occurrence and send the private colors of its atom tuple to the remaining entries of the chosen full tuple. For each coordinate of the selected semantic block, choose an admissible tuple witnessing the chosen certification coordinate of its image and use it to map the corresponding pin tuple. All nonshared pin entries are private, so these maps combine to a homomorphism from one enumerated lift to Q .

Conversely, let $\varphi: L \rightarrow Q$ be a palette homomorphism. Every pin tuple places its shared color in a singleton certification coordinate, so the global type map forces that image to have the prescribed type and certifies it in the chosen half. Consequently each atom tuple maps to a wholly certified record in its selected signal box, and its active pair belongs to the corresponding H_R^ε . The formal lift imposes all prescribed identifications of repeated query variables. If an atom has equal endpoint type, its two active coordinates are distinct. Their images are distinct either because Q is nondegenerate or by the second hypothesis on the selected signal copy (which holds in every intended palette because the selected typed relation is loopless). Hence the active pair is not removed by the diagonal exclusion in the definition of H_R^ε . The images therefore give an occurrence of $\mathbf{q}$. Other coincidences among images of distinct formal colors are harmless because query occurrences are not required to be injective. $\square$

Lemma 5.7. Let $(H, <)$ be an ordered K -graph, and let $((F_\ell, i_\ell))_{\ell=1}^r$ be a finite list. Each F_ℓ is a covered $(K-1)$ -face of H , repetitions are allowed, and $i_\ell \in [K]$. There exist pairwise distinct fresh vertices p_ℓ and an extension of the order such that, after adding all edges $F_\ell \cup \{p_\ell\}$, the face F_ℓ occurs in coordinate i_ℓ of its new edge. Every other facet of a new edge is private; two new edges share a covered face if and only if their attachment faces are equal.

Proof. For each ℓ , require p_ℓ to lie after the first $i_\ell - 1$ vertices of the ordered face F_ℓ and before all its remaining vertices. Choose an ambient gap after the $(i_\ell - 1)$ st vertex of F_ℓ and before the i_ℓ th vertex of F_ℓ , using an end gap when $i_\ell = 1$ or $i_\ell = K$. The requirements for different fresh vertices impose no order between those vertices, so all can be inserted simultaneously. When several vertices occupy the same ambient gap, order them arbitrarily. In the new edge, deleting p_ℓ leaves F_ℓ in coordinate i_ℓ . Every other facet contains the private vertex p_ℓ and is therefore new. Two distinct new edges have a common $(K-1)$ -face only when they contain the same old attachment face; in that case the common face is exactly that attachment face. $\square$

Lemma 5.8. *Every enumerated pinned full-record lift is isomorphic to the canonical palette of an effectively constructible finite ordered K -graph.*

Proof. Recall the canonical-coordinate convention. If an ordered K -edge is $e = \{v_1 < \dots < v_K\}$, then coordinate i of its canonical tuple is the facet $e \setminus \{v_i\}$. We first realize the atom tuples in the positive half. Take an ordered core $C_0 := \{c_1 < \dots < c_{K-2}\}$. For every query variable x of atomic type X , add one rim vertex z_x in the gap after c_{N+s_X-1} . Rims in the same gap may be ordered arbitrarily, except that for the unique possible same-type atom $R(x, y)$ we impose $z_y < z_x$. The manifest has at most one same-type atom, so these order requirements are compatible. For an atom $\alpha = R(x, y)$, add $E_\alpha := C_0 \cup \{z_x, z_y\}$. If $X \neq Y$, then within E_α

$$\text{rk}(z_x) = N + s_X + \mathbf{1}_{\{s_Y < s_X\}}, \quad \text{rk}(z_y) = N + s_Y + \mathbf{1}_{\{s_X < s_Y\}}.$$

The face carrying x is $E_\alpha \setminus \{z_y\}$, so its coordinate is $N + j_X(R)$; similarly the face carrying y has coordinate $N + j_Y(R)$. If $X = Y$, the imposed order gives the consecutive coordinates $N + s_X$ and $N + s_X + 1$ in the directed order of the atom.

We now classify all covered faces of the atom K -graph. For a query variable x , put $F_x := C_0 \cup \{z_x\}$. This face occurs in exactly the atom edges whose atom contains x , and it represents the formal color c_x . Every other facet of an atom edge is assigned to the private color occupying the same coordinate of T_α . If $\alpha \neq \alpha'$, then

$$|E_\alpha \cap E_{\alpha'}| = K - 2 + |\{x, y\} \cap \{x', y'\}|.$$

The simple-incidence conclusion of Lemma 4.4 excludes two distinct atoms on the same unordered variable pair. Hence two atom edges share a covered face exactly when their atoms share one query variable, and the common face is the corresponding F_x . Thus the canonical records of the atom K -graph are exactly the formal atom tuples, with all and only the prescribed color identifications.

For every pin tuple $P_{\alpha,i}$, attach a leaf edge to the facet of E_α representing $c_{\alpha,i}$, at the chosen certification coordinate $k(\alpha, i)$, using Lemma 5.7. Map its other facets to the private colors of $P_{\alpha,i}$ in their coordinate order. The face classification in that lemma shows that an atom edge and a pin edge share only the intended attachment face. Two pin edges share a covered face only when their formal pin tuples share the same attached color, and every other pin facet is private. Consequently the canonical palette of the resulting ordered K -graph is isomorphic, tuple by tuple and color by color, to the formal pinned lift.

For a lift in the negative half, perform the same construction in normalized coordinates and reverse the entire vertex order. Reversing the order replaces each coordinate i by $K + 1 - i$ and realizes the selected negative-half record. Every step is a finite explicit operation, so the construction is effective. $\square$

Theorem 5.9. *There exists an effectively constructible finite family $\mathcal{F}$ of K -graphs with the following properties.*

- (i) *Every nondegenerate $\mathcal{F}$ -deficient palette Q admits a homomorphism $Q \rightarrow P_{\text{type}}$. For every such homomorphism and each half, the relations extracted on the certified colors relative to the corresponding global type map are query-free.*
- (ii) *Every intended palindromic palette built from a query-free typed record is $\mathcal{F}$ -deficient.*

Proof. For a color c in a palette Q , let $I_Q(c)$ be the set of coordinates in which it occurs, with the convention $\bigcap_{i \in \emptyset} \mathcal{A}_i = \mathcal{T}$. Thus a color unused by every admissible tuple may be assigned any type. A type map exists exactly when

$$\bigcap_{i \in I_Q(c)} \mathcal{A}_i \neq \emptyset \tag{5.11}$$

for every c . For every inclusion-minimal set $I \subseteq [K]$ with $\bigcap_{i \in I} \mathcal{A}_i = \emptyset$, take one ordered $(K-1)$ -set F and, for each $i \in I$, add a fresh vertex p_i in a position for which F is the i th facet of $F \cup \{p_i\}$. The resulting ordered canonical star has all its edges sharing exactly the face F , with that face occurring at coordinate i in the edge indexed by i , and every other face private. If (5.11) fails in a nondegenerate target, choose one witnessing admissible tuple for each coordinate in the minimal bad set; these tuples map the corresponding star to the target. Excluding all stars forces a global type map. Choosing, for each color c , a type in $\bigcap_{i \in I_Q(c)} \mathcal{A}_i$ gives a palette homomorphism $Q \rightarrow P_{\text{type}}$.

For every expanded query, every half, and every enumerated choice of signal copies and certification coordinates in the pinned lift, include the ordered K -graph supplied by [Lemma 5.8](#). Apply [Lemma 5.4](#) to every shell star and query lift and let $\mathcal{F}$ be the resulting finite family of K -graphs.

If a nondegenerate $\mathcal{F}$ -deficient palette had no type map, a shell star would map to it and the corresponding outer-order K -graph would be colorable. After fixing an arbitrary global type map, a query occurrence in either half would, by [Lemma 5.6](#), give a pinned lift mapping to the target, and hence another forbidden outer-order K -graph would be colorable. Since the global type map was arbitrary, these two obstruction arguments prove part (i).

For (ii), an intended palette maps to P_{type} , so no shell star maps to it. Because $\text{rev}(P_{\text{type}}) = P_{\text{type}}$, no reversed shell star maps either. Query-freeness and [Lemma 5.6](#) exclude every canonical pinned lift; the active entries of a same-type signal relation cannot collapse because that relation is loopless. The intended full palette is palindromic, so a reversed lift mapping to it would, after reversing coordinates, give the unreversed lift mapping to the same palette. Both branches of [Lemma 5.4](#) are therefore absent, and the intended palette is $\mathcal{F}$ -deficient. $\square$

5.4 The exact density formula

We can now compare an arbitrary deficient palette with the intended homogeneous palettes. The certification coordinates first force the type masses, and the semantic coordinates then recover the positive relational score. The only general fact needed for the comparison is that the density of a relation is at most the product of the densities of its coordinate projections.

For a finite coordinate set I , a relation $A \subseteq C^I$, and a probability measure μ on C , write $d_\mu(A) := \sum_{a \in A} \prod_{i \in I} \mu(a_i)$. If $I = I_1 \sqcup \dots \sqcup I_t$, then

$$A \subseteq \prod_{j=1}^t \text{proj}_{I_j} A, \quad d_\mu(A) \leq \prod_{j=1}^t d_\mu(\text{proj}_{I_j} A). \quad (5.12)$$

For a palette Q , we abbreviate $d_\mu(Q) := d_\mu(A(Q))$. Let I_+, I_- be the two half-coordinate sets of the homogeneous construction and define $Q_\varepsilon := \text{proj}_{I_\varepsilon} A(Q)$ for $\varepsilon \in \{+, -\}$, where the coordinate order of Q_- is reversed whenever it is normalized as a positive half.

Lemma 5.10. *Fix one normalized half ε of a nondegenerate palette with a global type map, a probability measure μ , certified color sets $C_v^{\varepsilon,0}$, and masses $u_v^\varepsilon := \mu(C_v^{\varepsilon,0})$. Write $d_{\text{sem}}^\varepsilon$ for the density of the semantic projection and use the extracted relations H_R^ε defined before [Lemma 5.6](#). The following statements hold.*

- (i) *the certification projection has density at most $\Theta(\mathbf{u}^\varepsilon)$;*
- (ii) *if every $u_v^\varepsilon > 0$ and $\hat{\mu}_\varepsilon(c) := p_v \mu(c) / u_v^\varepsilon$ for $c \in C_v^{\varepsilon,0}$, then for every $R: X_R \rightarrow Y_R$,*

$$h_{\hat{\mu}_\varepsilon}(H_R^\varepsilon) = p_{X_R} p_{Y_R} \frac{h_\mu(H_R^\varepsilon)}{u_{X_R}^\varepsilon u_{Y_R}^\varepsilon},$$

with the same formula, and the diagonal excluded, when $X_R = Y_R$;

- (iii) *the semantic projection satisfies*

$$d_{\text{sem}}^\varepsilon \leq 1 - M\Theta(\mathbf{u}^\varepsilon) + L_0\Theta(\mathbf{u}^\varepsilon) \sum_{R \in \mathcal{R}} c_R h_{\hat{\mu}_\varepsilon}(H_R^\varepsilon); \quad (5.13)$$

- (iv) *for an intended typed record with exact type masses $\mathbf{p}$, all nonsignal boxes allowed, and relation R placed in every signal copy for R , the certification density is θ_* and the semantic density is exactly*

$$1 - M\theta_* + L_0\theta_* \sum_{R \in \mathcal{R}} c_R h(R). \quad (5.14)$$

Proof. The certification block is the Cartesian product of d_v singleton-type coordinates for each v , so restricting every coordinate to certified colors gives (i). For (ii), the normalization multiplies the weight of an ordered X_R, Y_R pair by $p_{X_R} p_{Y_R} / (u_{X_R}^\varepsilon u_{Y_R}^\varepsilon)$; looplessness removes the diagonal in the same-type case.

For (iii), partition the semantic universe by its length- N type word. The M signal words are pairwise distinct, and the wholly certified part of each signal box has product mass $\Theta(\mathbf{u}^\varepsilon)$. Hence all nonsignal boxes together with the portions of signal boxes containing an uncertified coordinate have total mass at most $1 - M\Theta(\mathbf{u}^\varepsilon)$. In one signal copy for $R: X_R \rightarrow Y_R$, fixing an active pair leaves a product mass $\Theta(\mathbf{u}^\varepsilon)/(u_{X_R}^\varepsilon u_{Y_R}^\varepsilon)$ in the remaining coordinates. The allowed active pairs of that copy form a subset of H_R^ε , so its wholly certified contribution is at most this factor times $h_\mu(H_R^\varepsilon)$. Summing the $m_R = L_0 c_R p_{X_R} p_{Y_R}$ copies and using (ii) gives (5.13).

For (iv), the type-word boxes partition the entire semantic universe because the type masses sum to one. Every signal word has mass θ_* , while one copy for R has allowed mass $\theta_* h(R)/(p_{X_R} p_{Y_R})$. Summing its m_R copies and allowing all nonsignal boxes gives (5.14). The certification block has product density θ_* by construction. $\square$

Theorem 5.11. *For the finite family $\mathcal{F}$ constructed in Theorem 5.9,*

$$\pi_{K-2}^{(K)}(\mathcal{F}) = f_\gamma(\theta_*)^2 = (\alpha_0 + \alpha_1 \gamma)^2,$$

where

$$\alpha_0 := \theta_* - \theta_*^2 (L_0 \mathcal{B}(\mathbf{p}_0) + M), \quad \alpha_1 := \frac{L_0}{2} p_Y^2 \theta_*^2 > 0. \quad (5.15)$$

Proof. We first prove the upper bound. By Lemma 2.3, it is enough to consider a nondegenerate $\mathcal{F}$ -deficient palette Q with an arbitrary probability measure μ . The shell obstructions give one global type map. Fix a half ε and reverse its coordinate order when $\varepsilon = -$. Let $C_v^{\varepsilon,0}$ be the set of ε -certified colors of global type v , and put $u_v^\varepsilon := \mu(C_v^{\varepsilon,0})$. Put $\mathbf{u}^\varepsilon := (u_v^\varepsilon)_{v \in \mathcal{T}}$. These sets are disjoint, so $\sum_v u_v^\varepsilon \leq 1$. By Lemma 5.10(i), the certification projection has density at most $\Theta(\mathbf{u}^\varepsilon)$. If one factor is zero, the half density is zero. Otherwise, for $c \in C_v^{\varepsilon,0}$, define $\hat{\mu}_\varepsilon(c) := p_v \mu(c)/u_v^\varepsilon$. The total $\hat{\mu}_\varepsilon$ -mass of type v is p_v . For $R: X_R \rightarrow Y_R$, use the extracted relation H_R^ε defined before Lemma 5.6. Parts (ii) and (iii) of Lemma 5.10 give

$$d_{\text{sem}}^\varepsilon \leq 1 - M\Theta(\mathbf{u}^\varepsilon) + L_0 \Theta(\mathbf{u}^\varepsilon) \sum_{R \in \mathcal{R}} c_R h_{\hat{\mu}_\varepsilon}(H_R^\varepsilon).$$

By Theorem 5.9(i), the extracted structure is query-free. Restrict its colors and the measure $\hat{\mu}_\varepsilon$ to the endpoint types $\mathcal{T}_0$. The resulting weighted $\mathcal{T}_0$ -record has type-mass vector $\mathbf{p}_0$, remains query-free, and has the same values of every $h_{\hat{\mu}_\varepsilon}(H_R^\varepsilon)$ because no relation is incident with a filler type. Repair this restricted record by Theorem 4.16; the repair preserves $\mathbf{p}_0$ and does not decrease the positive relation score. The source mass in the stopped-tree estimate is at most p_Y . Therefore

$$\mathcal{B}(\mathbf{p}_0) + \sum_R c_R h_{\hat{\mu}_\varepsilon}(H_R^\varepsilon) \leq \frac{1}{2} p_Y^2 \gamma - \Delta$$

for some $\Delta \geq 0$. Substitution gives

$$d_{\text{sem}}^\varepsilon \leq 1 + \Theta(\mathbf{u}^\varepsilon)(\beta(\gamma) - L_0 \Delta) \leq 1 + \beta(\gamma)\Theta(\mathbf{u}^\varepsilon).$$

By (5.12), the half relation is contained in the product of its certification and semantic projections, and thus

$$d_\mu(Q_\varepsilon) \leq f_\gamma(\Theta(\mathbf{u}^\varepsilon)).$$

Weighted AM–GM gives

$$\Theta(\mathbf{u}^\varepsilon)^{1/N} = \left(\prod_v p_v^{p_v} \right) \prod_v \left(\frac{u_v^\varepsilon}{p_v} \right)^{p_v} \leq \left(\prod_v p_v^{p_v} \right) \sum_v u_v^\varepsilon,$$

so $\Theta(\mathbf{u}^\varepsilon) \leq \theta_*$. By (5.6), f_γ is increasing on $[0, \theta_*]$, and hence each half has density at most $f_\gamma(\theta_*)$.

The forbidden family contains the pinned lifts for both halves. Thus the same query-free extraction applies to the reversed second half, even though its certified color sets may differ from those of the first half. Finally, the full allowed relation is contained in the Cartesian product of its two half projections. A second use of (5.12) gives

$$d_\mu(Q) \leq d_\mu(Q_+) d_\mu(Q_-) \leq f_\gamma(\theta_*)^2. \quad (5.16)$$

The bound holds for every μ , so the weighted palette transfer gives the upper bound for $\pi_{K-2}^{(K)}(\mathcal{F})$.

For the lower bound, fix a cutoff T_D . Apply [Lemma 4.15](#) to the restricted capacity vector $\mathbf{p}_0$. Adjoin finite color classes of the two relation-free filler types F_1, F_2 with total masses p_{F_1}, p_{F_2} and no incident relation edges. Adding the filler classes changes neither Φ nor the bridge defect or query-freeness, and it restores the full type-mass vector $\mathbf{p}$. Let $\mathfrak{R}_{D,n}$ be the resulting query-free record.

Let $P_{D,n}$ be the intended palette obtained by allowing every prescribed certification fiber, filling every nonsignal semantic box, and using the relation R of $\mathfrak{R}_{D,n}$ in the active coordinates of each signal copy for R . Let $\mu_{D,n}$ be the probability measure given by the record weights. By [Lemma 5.10\(iv\)](#), the certification-block density is exactly θ_* and

$$\begin{aligned} d_{\text{sem}}(D, n) &= 1 - M\theta_* + L_0\theta_* \sum_R c_R h_{\mathfrak{R}_{D,n}}(R) \\ &= 1 + \theta_* (L_0(\Phi(\mathfrak{R}_{D,n}) - \mathcal{B}(\mathbf{p}_0)) - M). \end{aligned}$$

By [Lemma 4.15](#), the semantic density tends to $1 + \beta(g_D)\theta_*$. The intended base half is the Cartesian product of its certification and semantic blocks, so its density tends to $f_{g_D}(\theta_*)$. The full intended palette is the Cartesian product of the base half and its coordinate reverse, so its density tends to $f_{g_D}(\theta_*)^2$.

By [Theorem 5.9\(ii\)](#), every $P_{D,n}$ is $\mathcal{F}$ -deficient. The weighted palette transfer therefore gives, for every D and n ,

$$\pi_{K-2}^{(K)}(\mathcal{F}) \geq d_{\mu_{D,n}}(P_{D,n}).$$

First let $n \rightarrow \infty$ to obtain $\pi_{K-2}^{(K)}(\mathcal{F}) \geq f_{g_D}(\theta_*)^2$, and then let $D \rightarrow \infty$. By [Proposition 3.8](#), $g_D \nearrow \gamma$, so this matches [\(5.16\)](#). Finally,

$$f_\gamma(\theta_*) = \theta_* + \theta_*^2 \left[L_0 \left(\frac{1}{2} p_Y^2 \gamma - \mathcal{B}(\mathbf{p}_0) \right) - M \right] = \alpha_0 + \alpha_1 \gamma,$$

which gives [\(5.15\)](#). □

Proof of [Theorem 1.1](#). Take $\mathcal{F}, K, \alpha_0$, and α_1 from [Theorem 5.11](#). That theorem gives $\pi_{K-2}^{(K)}(\mathcal{F}) = (\alpha_0 + \alpha_1 \gamma)^2$, and [\(5.15\)](#) gives $\alpha_1 > 0$. The value of K is the explicit integer in [\(5.3\)](#). The complete deterministic output algorithm, including the exact query manifest, rational certificate, greedy signal-word rule, and terminating outer-order search, is given in [Sections A](#) and [B](#). Hence the family is effectively constructible in the asserted literal sense. □

Proof of [Corollary 1.2](#). By [Corollary 3.6](#), the number $\gamma = G(1/100)$ is transcendental. If $(\alpha_0 + \alpha_1 \gamma)^2$ were algebraic, then $\alpha_0 + \alpha_1 \gamma$ would be algebraic over the field of algebraic numbers and hence algebraic over $\mathbb{Q}$. Since $\alpha_0, \alpha_1 \in \mathbb{Q}$ and $\alpha_1 > 0$, this would make γ algebraic, a contradiction. □

Proof of [Corollary 1.3](#). By [Corollary 2.4](#), the family in [Theorem 1.1](#) satisfies

$$\sup\{\Lambda(P) : P \text{ is a finite } \mathcal{F}\text{-deficient palette}\} = \pi_{K-2}^{(K)}(\mathcal{F}),$$

and the right side is transcendental by [Corollary 1.2](#). □

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Declaration on the use of AI

The authors used generative AI tools to discuss the organization of the proofs, check proof details, and improve the exposition.

A Finite certificates

This section separates the human-readable schemas from their complete finite expansion. The complete machine-readable signature and query expansion, together with the rational homogeneous-parameter certificate, are supplied in the ancillary files

manifest_generator.py, query_manifest.json,
relation_signature.tsv, construction_certificate.json,
and manifest_audit.txt.

The Python program uses exact integer and rational arithmetic, generates the complete lists, and checks every structural assertion used in [Lemma 4.4](#) and [Proposition 4.6](#). These files do not enumerate the signal words or the final hypergraphs; [Appendix B](#) gives terminating deterministic rules for both.

A.1 Signature and exact query counts

The 52 atomic types and 322 relation symbols have the census $52 = 1 + 3 + 12 + 4 \cdot 9$ and $322 = 1 + 12 + 30 + 15 + 30 + 27 + 54 + 153$. The summands on the right count, respectively, the type and relation classes listed in the proof of [Proposition 4.6](#).

query category	number of instances
package functionality	486
package injectivity	513
pairing functionality	42
pairing injectivity	240
scored-edge incoming uniqueness	96
scored-edge outgoing functionality	60
source functionality	2
relator zero/nonzero mismatch	27
relator nonzero-endpoint mismatch	120
ideal-word query	3
nonzero injectivity (zero_injectivity)	180
zero-edge compatibility	30
co-reference compatibility	600
I – E complement	30
sibling complement	27
total	2456

Table 1: Exact expansion count of the finite query manifest.

schema	nonreserve atoms	possible same-type atom	enforced property
outgoing conflict	$R(x, y), R(x, y')$	a reserve on y, y' when their atomic types agree	functionality of R
incoming conflict	$R(x, y), R(x', y)$	a reserve on x, x' when their atomic types agree	injectivity of R
source conflict	$S(y, o), S(y, o')$	$Q_{V_0, *}^{\text{sq}}(o, o')$	left-functionality of S
relator zero mismatch	$Z_{i, w, q}(x)$ and $\text{NZ}_{i, w'}(x, z)$	none	the two relator words vanish together
relator endpoint mismatch	$\text{NZ}_{i, w}(x, z)$ and $\text{NZ}_{i, w'}(x, z')$	a reserve on z, z'	equality of nonzero endpoints
ideal word	$\text{NZ}_{i, acc}(x, z)$	none	$R_c R_c R_a = 0$

Table 2 (continued)

schema	nonreserve atoms	possible same-type atom	enforced property
nonzero	two one-letter nonzero paths with a	a reserve on their	nonzero injectivity of
injectivity	common image	sources	R_s
zero edge	the four nonreserve atoms I, E, A, T in the zero-edge template	none	a scored child cannot certify a zero action
co-reference	the five nonreserve atoms I, E, I, A, T in (4.8)	a chosen $N \in \mathcal{N}_{U_j}$ on (z, u') , in either orientation	the semantic and scored children agree
$I-E$	$I(u, v), E_s(v, w)$	$C^{IE}(u, w)$	disjointness from the $I-E$ path relation
complement			
sibling	$E_s(v, w), E_t(v, w')$	$C^{EE}(w, w')$	disjointness from the sibling relation
complement			

Table 2: Human-readable schema table for the finite query manifest. Each row is expanded over the finite phase, channel, word, reserve, path-position, and orientation sets defined in Section 4.1; the exact expanded instances are in `query_manifest.json`.

The generator verifies the following assertions independently for each of the 2456 JSON records.

- (i) every relation symbol occurs with its declared ordered endpoint types;
- (ii) no atom is a formal loop $R(x, x)$;
- (iii) no two atoms use the same unordered variable pair;
- (iv) at most one atom has equal endpoint types, and it is a reserve atom;
- (v) every variable has one well-defined atomic type;
- (vi) the numbers of atoms and variables are at most 11.

It also recomputes the signature census, every category count in Table 1, and the 42 local-cell schemas. The audit file ends with the assertion that all exact-arithmetic and structural tests passed. The SHA-256 identifiers of the two complete finite data files are

```

query_manifest.json
6ec364ea1470dbfcbf3371021f31924323766273f839d24c6a5be74cd5f47a7e
relation_signature.tsv
9ecbfdc291aec84ac4c41ebf445dd9d17e08a6b9b1aa1a1d7f63908840f619ad

```

A.2 Local-cell charge allocation

Table 3 summarizes the four cases in the proof of Lemma 4.11.

case	condition	assigned atomic charges	resulting lower bound
large prefix error missing A -bridge	$\sum_{1 \leq h < j} \varepsilon_h^2 \geq W_{j-1}^2/16$ prefix error below threshold and the next A -edge is absent	$\sum_{1 \leq h < j} \delta_h$ unmatched-domain A -charge	$W_{j-1}^2/64$ $r_{s_j}^2 W_{j-1}^2/8$
missing T -bridge	prefix error below threshold, the A -edge is present, and the required T -edge is absent	matched A -edge charge plus unmatched-domain T -charge	$r_{s_j}^2 W_{j-1}^2/16$
zero target	a required bridge reaches a zero type	no defect charge; the corresponding zero-edge or zero/nonzero query occurs	excluded in a query-free structure

Table 3: Charge allocation for a concrete local cell. Here j is the first missing generator step, W_{j-1} is the ideal ratio weight at the preceding state, and δ_h is the sum of the matched A - and T -edge charges in the h th realized transition.

For the overlap count in Lemma 4.11, the fifteen roles are as follows.

role family	number of roles	indices
matched A - or T -edge in a realized prefix	6	$1 \leq h < j \leq 3$, with the package choice A or T
unmatched A -domain charge at the first missing step	3	$j \in \{1, 2, 3\}$
the two charges used for a missing T -bridge	6	$j \in \{1, 2, 3\}$ and the charge choice

Fixing an atomic charge, a schema, and one of these roles recovers at most one concrete cell by the matching and nonzero injectivity argument in the proof. This uniqueness is the origin of the factor $15|\mathcal{L}_{\text{cell}}|$.

B Effective construction

This section gives a deterministic specification of the finite output in Theorem 1.1. It is not an additional existence argument. Every step is the canonical finite choice used in the body of the proof.

B.1 Exact numerical certificate

The constants used before homogenization are

$$\begin{aligned}
c_{\text{cell}} &= \frac{1}{64,000,000,000,000}, & C_{\text{cell}} &= 630, \\
C_I &= \frac{57,335,040,026,661,119,200}{6,967,089}, & C_B &= \frac{280,970,363,520,026,661,119,200}{6,967,089}, \\
M_0 &= \frac{70,242,590,880,006,679,213,978}{6,967,089}.
\end{aligned}$$

The homogeneous constants and their integer outputs are

$$\begin{aligned}
U_* &= \frac{100}{79}, & V_* &= \frac{20000}{6241}, & p_Y &= \frac{6241}{1810514}, \\
L_0 &= 72,271,663,407,397,359, & N &= 3,621,028, & K &= 14,484,112.
\end{aligned}$$

$$M = 8,527,079,511,467,783,033,168,342,073,204.$$

The multiplicity vector $d_v = Np_v$ is displayed in (5.7). The complete 322-entry vector $(m_R)_{R \in \mathcal{R}}$ is stored in `construction_certificate.json`; its minimum and maximum entries are 17,638,200,000 and 221,966,587,180,821,106,316,170,480,000. The same certificate contains the exact values in (5.8)–(5.9) and the bounds $M < 2^{103}$, $|\beta| < 2^{102}$, and $\theta_* \leq 2^{-N}$. Its SHA-256 identifier is

$$\text{f193c435e2d65c4cf78211705c9e0f11b0e79a24213a3850ad58f34f368593eb.}$$

B.2 Deterministic output algorithm

Fix the deterministic orders stipulated before [Lemma 4.4](#). The following algorithm outputs a concrete encoding of the family $\mathcal{F}$.

- Step 1.** Enumerate the 52 atomic types and 322 positive relation symbols. Expand all query schemas canonically and run the six structural tests listed in [Section A](#). Reject the computation unless the census is exactly 2456.
- Step 2.** Assign the rational masses in [Construction 5.2](#). Compute every $m_R = L_0 c_R p_{X_R} p_{Y_R}$ and verify that it is a positive integer and that their sum is the displayed value of M . Compute every $d_v = N p_v$ and verify [\(5.7\)](#).
- Step 3.** Order the endpoint types as $X_0, \dots, X_{51}$, use $s_{X_q} = 2 + 3q$, and order the M signal-copy requests lexicographically. For each request, place the prescribed active entries, fill the remaining nonfiller entries in the least currently unassigned coordinates in the displayed type order as in [Lemma 5.3](#), and then choose the lexicographically first unused assignment of F_1 and F_2 to the remaining coordinates. The completion bound in that lemma proves that this greedy search never fails.
- Step 4.** Enumerate all inclusion-minimal bad coordinate sets for the shell stars. For every expanded query, both halves, every allowed choice of signal copies, and every allowed choice of certification coordinates, construct the formal pinned lift and then its ordered common-core realization from [Lemma 5.8](#).
- Step 5.** For every ordered gadget $(J, <)$ from Step 4, delete its isolated vertices and execute the terminating finite search in [Lemma 5.4](#). Enumerate the ground-set size, all uniform block families, and all total orders until a linear design with a monotone block in every order is found. Form $\mathfrak{D}(J)$.
- Step 6.** For each K -graph $\mathfrak{D}(J)$ obtained in Step 5, examine all vertex labelings by an initial interval of positive integers and choose the lexicographically least sorted edge list. Sort these canonical adjacency-list encodings lexicographically and delete duplicates.

Every loop ranges over an explicitly bounded finite set except the increasing search in Step 5; [Lemma 5.4](#) proves that the latter terminates. Thus the algorithm is a total deterministic construction, rather than only a recursively enumerable specification.

Proposition B.1. *The algorithm above outputs a family satisfying*

$$|\mathcal{F}| \leq 2^K + 2 \cdot 2456 M^{11} N^{11N}. \quad (\text{B.1})$$

Moreover, define $d(m)$ to be the least ground-set size of a linear m -uniform design having a monotone block for every total order, as in [Lemma 5.4](#), and put

$$m_* := 15N + 9 = 54,315,429, \quad D_* := \max_{1 \leq m \leq m_*} d(m).$$

Every member of $\mathcal{F}$ has at most D_* vertices. Both bounds are effective integers.

Proof. There is at most one shell star for each subset of $[K]$, hence at most 2^K shell gadgets. A query has at most 11 atoms. For each atom there are at most M choices of signal copy and at most N^N choices of the N certification coordinates. Multiplying by the 2456 queries and two halves gives the second term in [\(B.1\)](#).

A shell star has at most $2K - 1$ vertices. A common-core realization of a pinned lift has at most $K - 2$ core vertices, 11 query-variable vertices, and $11N$ private pin vertices. Its order is therefore at most $K - 2 + 11 + 11N = 15N + 9 = m_*$. The outer-ordering construction for a gadget on m nonisolated vertices uses the ground set of its design and hence has $d(m) \leq D_*$ vertices. Finally, each $d(m)$ and the finite maximum D_* are computable by the terminating enumeration in [Lemma 5.4](#). $\square$

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